COUNTRY PROFILES

Nuclear Power in India

UPDATED TUESDAY, 20 JANUARY 2026

- India has a largely indigenous nuclear power programme.
- The Indian government is committed to growing its nuclear power capacity as part of its massive infrastructure development programme.
- The government has set an ambitious target to grow nuclear capacity to at least 100 GWe by 2047.
- Because India is outside the Nuclear Non-Proliferation Treaty due to its weapons programme, it was for 34

Privacy - Terms

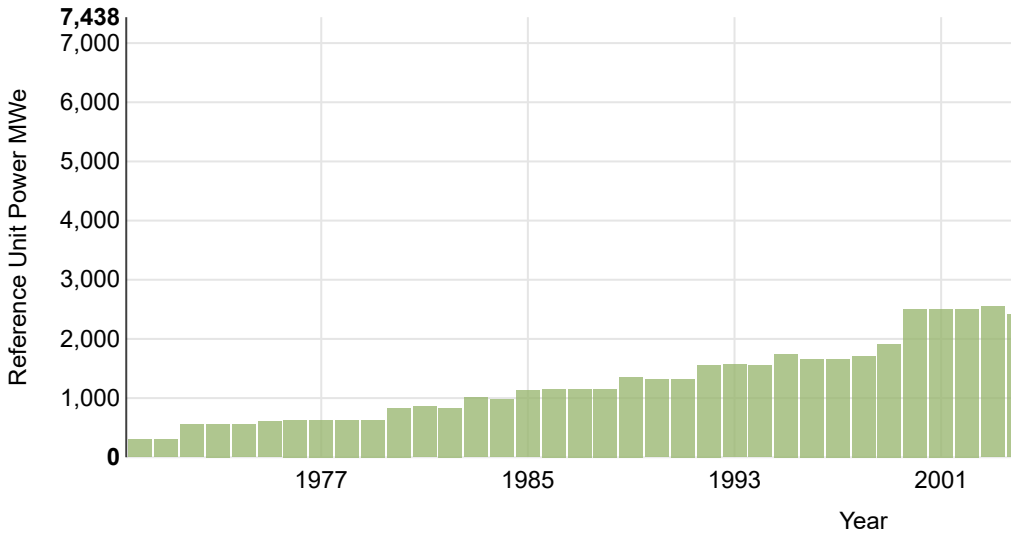
years largely excluded from trade in nuclear plant and materials, which hampered its development of civil nuclear energy until 2009.

- Since 2010, a fundamental incompatibility between India's civil liability law and international conventions limits foreign technology provision.

- Due to earlier trade bans and lack of indigenous uranium, India has uniquely been developing a nuclear fuel cycle to exploit its reserves of thorium.

24	8	1
OPERABLE REACTORS	REACTORS UNDER CONSTRUCTION	REACTORS SHUTDOWN
7,943 MWe	6,028 MWe	135 MWe

Operable nuclear power capacity



Electricity sector

Total generation (in 2023): 1987 TWh

Generation mix: coal 1479 TWh (74%); hydro 144 TWh (7%); solar 119 TWh (6%); wind 93.5 TWh (5%); natural gas 58.8 TWh (3%); nuclear 47.9 TWh (2%); biofuels & waste 41.4 TWh (2%); oil 4.2 TWh

Import/export balance: 6.2 TWh net export (7.6 TWh imports; 13.8 exports)

Total consumption: 1574 TWh

Per capita consumption: c. 1100 kWh in 2023

Source: International Energy Agency and The World Bank. Data for year 2023.

Energy policy

India has a population of over 1.4 billion and a fast-growing economy. Energy demand is expected to grow more in India than any other country over the next decade. India's priorities are economic growth and to alleviate poverty. As such, coal is to remain the largest source of energy supply until at least the middle of the century. At COP26 India pledged to cut its emissions to net zero by 2070 – the first time it made such a commitment.

Nuclear energy plays an important role in the country's long-term energy strategy. The country has developed an indigenous nuclear power programme, based on a fuel cycle which aims to utilize the country's vast thorium reserves.

In March 2018, the government stated that nuclear capacity would be about 22.5 GWe by the year 2031[c], well below earlier targets (see below). This revised target was reaffirmed by Minister of State Jitendra Singh in December 2022[d], and by the Department of Atomic Energy (DAE) in February 2025.

Viksit Bharat 2047

In December 2023 the government launched *Viksit Bharat*, its strategy to make India a developed nation by 2047 (*i.e.* 100 years after independence). In July 2024 the country's 2024-25 budget specified setting up Bharat Small Reactors*. In December 2024 NPCIL issued a request for proposals (RFP) from Indian energy users to finance and build a proposed fleet of Bharat Small Reactors. The initial deadline was 31 March 2025, but this has since been extended several times. In October 2025 it was

extended to 31 March 2026. Six companies at that time had submitted documents in response to the RFP: Hindalco Industries; Jindal Steel & Power; Tata Power; Reliance Industries; JSW Energy; and Adani Power. The six companies identified 16 prospective sites across six states, including five in Gujarat, four in Madhya Pradesh, three in Odisha, two in Andhra Pradesh, and one each in Jharkhand and Chhattisgarh.

Earlier in February 2025 Minister of Finance Nirmala Sitharaman announced the Nuclear Energy Mission for Viksit Bharat, and promised federal funds to develop at least five Indian-designed SMRs to be operational by 2033, as well as amendments to Indian legislation to encourage private sector participation, in the budget for 2025-2026. In the same budget she said that the "development of at least 100 GW of nuclear energy by 2047 is essential for our energy transition efforts." New reactors would be built at existing nuclear power plants, greenfield and brownfield sites and to directly replace coal-fired power units.

* Bharat Small Reactors, first announced in August 2023 by Minister of State Jitendra Singh, are expected to be closely based on the existing 220 MWe PHWR technology India has built since the 1980s. In August 2025 it was announced that BARC was developing three reactor designs: The 200 MWe 'Bharat Small Modular Reactor', a 50 MWe SMR and a 5 MWt high temperature gas cooled reactor that could be coupled with chemical processes such as the production of hydrogen. 'In principle' approval was given for demonstration of these designs at sites owned by the DAE sites in collaboration with Nuclear Power Corporation of India Ltd (NPCIL).

Also in February 2025 India and France signed a letter of intent to partner on advanced modular and small modular reactors. In the same month India and the USA agreed to fully realize the US-India 123 Civil Nuclear Agreement by moving forward with plans to build US-designed reactors in India.

In May 2025 the country's Minister for Power set out the steps needed to scale the country's nuclear energy capacity:

- Amending the Atomic Energy Act, 1962 and Civil Liability for Nuclear Damage Act, 2010 to enable broader participation by private and state sectors
- Strengthening public perception and enhancing awareness about nuclear energy's safety and benefits
- Facilitating faster land acquisition through brownfield

expansions and repurposing retired thermal sites

- Streamlining regulatory approval processes to reduce project timelines
- Introducing tax concessions, green power classification, and long-term financing to ensure competitive nuclear tariffs
- Diversifying technology choices through competitive bidding and promoting indigenous manufacturing under Make in India
- Securing diversified uranium fuel sources and expanding the vendor base for specialized nuclear equipment
- Building skilled manpower capacity by strengthening nuclear education and training infrastructure

In September 2025 the chair of India's Central Electricity Authority said that the government had identified four retiring coal plant sites that could be used for siting new nuclear reactors.

In December 2025 the Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Bill passed both houses of India's parliament. The legislation replaces the 1962 Atomic Energy Act and the 2010 Civil Liability for Nuclear Damage Act with a single, comprehensive law. See section on Nuclear liability below.

Nuclear power industry

Reactors operating in India

Name	Model	Reactor Type	Reference Unit Power (MWe)	Grid Connection
Kaiga 1	Horizontal Pressure Tube type	PHWR	202	2000-10
Kaiga 2	Horizontal Pressure Tube type	PHWR	202	1999-12
Kaiga 3	Horizontal Pressure Tube type	PHWR	202	2007-04
Kaiga 4	Horizontal Pressure Tube type	PHWR	202	2011-01

Name	Model	Reactor Type	Reference Unit Power (MWe)	Grid Connection
Kakrapar 1	Horizontal Pressure Tube type	PHWR	202	1992-11
Kakrapar 2	Horizontal Pressure Tube type	PHWR	202	1995-03
Kakrapar 3	PHWR-700	PHWR	630	2021-01
Kakrapar 4	PHWR-700	PHWR	630	2024-02
Kudankulam 1	VVER V-412	PWR	932	2013-10
Kudankulam 2	VVER V-412	PWR	932	2016-08
Madras 1	Horizontal Pressure Tube type	PHWR	205	1983-07
Madras 2	Horizontal Pressure Tube type	PHWR	205	1985-09
Narora 1	Horizontal Pressure Tube type	PHWR	202	1989-07
Narora 2	Horizontal Pressure Tube type	PHWR	202	1992-01
Rajasthan 2	Horizontal Pressure Tube type	PHWR	187	1980-11
Rajasthan 3	Horizontal Pressure Tube type	PHWR	202	2000-03
Rajasthan 4	Horizontal Pressure Tube type	PHWR	202	2000-11
Rajasthan 5	Horizontal Pressure Tube type	PHWR	202	2009-12
Rajasthan 6	Horizontal Pressure Tube type	PHWR	202	2010-03
Rajasthan 7	Horizontal Pressure Tube type	PHWR	630	2025-03
Tarapur 1	BWR-1 (Mark 2)	BWR	150	1969-04
Tarapur 2	BWR-1 (Mark 2)	BWR	150	1969-05
Tarapur 3	Horizontal Pressure Tube Type	PHWR	490	2006-06
Tarapur 4	Horizontal Pressure Tube Type	PHWR	378	2005-06

Reactors under construction in India

Reactor Name	Model	Reactor Type	Gross Capacity	Construction Start
Kaiga 5	Horizontal Pressure Tube type	PHWR	700	2026-03-01
Kaiga 6	Horizontal Pressure Tube type	PHWR	700	2026-03-01
Kudankulam 3	VVER V-412	PWR	1000	2017-06-29
Kudankulam 4	VVER V-412	PWR	1000	2017-10-23
Kudankulam 5	VVER V-412	PWR	1000	2021-06-29
Kudankulam 6	VVER V-412	PWR	1000	2021-12-20
PFBR	Prototype	FBR	500	2004-10-23
Rajasthan 8	Horizontal Pressure Tube type	PHWR	700	2011-09-30

Nuclear capacity targets

The target since about 2004 was for nuclear power to provide 20 GWe by 2020, but in 2007 the prime minister referred to this as "modest" and capable of being "doubled with the opening up of international cooperation." In June 2009 NPCIL said it aimed for 60 GWe nuclear by 2032, including 40 GWe of PWR capacity and 7 GWe of new PHWR capacity, all fuelled by imported uranium. This 2032 target was reiterated late in 2010 and increased to 63 GWe in 2011. But in December 2011 parliament was told that more realistic targets were 14,600 MWe by 2020-21 and 27,500 MWe by 2032.*

* "The XII Plan [2012-17] proposals ..... envisage start of work on eight indigenous 700 MW pressurized heavy water reactors (PHWRs), two 500 MW fast breeder reactors (FBRs), one 300 MW advanced heavy water reactor (AHWR) and eight light water reactors of 1000 MW or higher

capacity with foreign technical cooperation. These nuclear power reactors are expected to be completed progressively in the XIII and XIV Plans." The 16 PHWRS and LWRs were expected to cost $40 billion. The eight 700 MWe PHWRs were to be built at Kaiga in Karnataka, Gorakhpur in Haryana's Fatehabad District, Banswara in Rajasthan, and Chutka in Madhya Pradesh.

In July 2014 the new prime minister urged the DAE to triple the nuclear capacity to 17 GWe by 2024. He praised "India's self-reliance in the nuclear fuel cycle and the commercial success of the indigenous reactors." He also emphasized the importance of maintaining the commercial viability and competitiveness of nuclear energy compared with other clean energy sources. In March 2017 parliament was told that the 14.6 GWe target of nuclear capacity by 2024 was maintained, relative to 6.7 GWe (gross) grid-connected then.

In May 2017 the cabinet approved ten 700 MWe PHWRs as a "fully homegrown initiative" with likely manufacturing orders to Indian industry of about INR 700 billion ($11 billion). These would be at four sites identified in 2015, but without a timeline being specified. The prime minister said it would help transform the domestic nuclear industry, which appeared to suggest lower expectations of establishing new nuclear plants with Western technology from Areva, GE Hitachi, and Westinghouse. No mention was made of the other elements of the 12th five-year plan for 2012-17, *i.e.* the Western LWRs which were originally intended to accelerate new capacity additions, and also two FBRs and one AHWR. Parliament fully supported the announcement.

In March 2018, the government stated that nuclear capacity would fall well short of its 63 GWe target outlined in the 12th five-year plan for 2012-17, and that total nuclear capacity is likely to be about 22.5 GWe by the year 2031[c]. These plans have since been reaffirmed.

In February 2025 Minister of Finance Nirmala Sitharaman announced that the "Development of at least 100 GW of nuclear energy by 2047 is essential for our energy transition efforts" (see Viksit Bharat 2047 section above).

Other energy information for India: US Energy Information Administration Analysis Brief on India

Industry development

Nuclear power for civil use is well established in India. Since building the two small boiling water reactors at Tarapur in the 1960s, its civil nuclear strategy has been directed towards complete independence in the nuclear fuel cycle, necessary because it is excluded from the 1970 Nuclear Non-Proliferation Treaty (NPT) due to it acquiring nuclear weapons capability after 1970. (Those five countries doing so before 1970 were accorded the status of Nuclear Weapons States under the NPT.)

As a result, India's nuclear power programme has proceeded largely without fuel or technological assistance from other countries (but see later section). The pressurized heavy-water reactor (PHWR) design was adopted in 1964, since it required less natural uranium than the BWRs, needed no enrichment, and could be built with the country's engineering capacity at that time – pressure tubes rather than a heavy pressure vessel being involved. Its power reactors to the mid-1990s had some of the world's lowest capacity factors, reflecting the technical difficulties of the country's isolation, but rose impressively from 60% in 1995 to 85% in 2001-02. Then in 2008-10 the load factors dropped due to shortage of uranium fuel.

India's nuclear energy self-sufficiency extended from uranium exploration and mining through fuel fabrication, heavy water production, reactor design and construction, to reprocessing and waste management. It has a small fast breeder reactor and is building a much larger one. It is also developing technology to utilize its abundant resources of thorium as a nuclear fuel.

The Atomic Energy Establishment was set up at Trombay, near Mumbai, in 1957 and renamed as Bhabha Atomic Research Centre (BARC) ten years later. Plans for building the first Pressurized Heavy Water Reactor (PHWR) were finalized in 1964, and this prototype – Rajasthan 1, which had Canada's Douglas Point reactor as a reference unit, was built as a collaborative venture between Atomic Energy of Canada Ltd (AECL) and NPCIL. It started up in 1972 and was duplicated Subsequent indigenous PHWR development has been based on these units, though several stages of evolution can be identified: PHWRs with dousing and single containment at Rajasthan 1-2, PHWRs with suppression pool and partial double containment at Madras, and later standardized PHWRs from Narora onwards having double containment, suppression pool, and calandria filled with heavy water, housed in a water-filled calandria vault.

The Indian Atomic Energy Commission (AEC) is the main policy body.

The Nuclear Power Corporation of India Ltd (NPCIL) is responsible for design, construction, commissioning and operation of thermal nuclear power plants. At the start of 2010 it said it had enough cash on hand for 10,000 MWe of new plant. Its funding model is 70% equity and 30% debt financing. However, it is aiming to involve other public sector and private corporations in future nuclear power expansion, notably National Thermal Power Corporation (NTPC) – see subsection below. NTPC is very much larger than NPCIL and sees itself as the main power producer. NTPC is largely government-owned. The 1962 Atomic Energy Act prohibits private control of nuclear power generation, and 2016 amendments allowing public sector joint ventures do not extend to private sector companies, nor allow direct foreign investment in nuclear power, apart from the supply chain.

Specific reactors

In July 2017, eight reactors – 2400 MWe (gross) – of nuclear capacity was fuelled by indigenous uranium and being operated close to their rated capacity. The 14 units (4380 MWe gross) under safeguards were operating on imported uranium at rated capacity.

The two **Tarapur** 150 MWe boiling water reactors (BWRs) built by GE on a turnkey contract before the advent of the Nuclear Non-Proliferation Treaty were originally 200 MWe. They were downrated due to recurrent problems but have run reasonably well since. They have been using imported enriched uranium (from France and China in 1980-90s and Russia since 2001) and are under International Atomic Energy Agency (IAEA) safeguards. However, late in 2004 Russia deferred to the Nuclear Suppliers' Group and declined to supply further uranium for them. They underwent six months' refurbishment over 2005-06, and in March 2006 Russia agreed to resume fuel supply. In December 2008 a $700 million contract with Rosatom was announced for continued uranium supply to them. In 2015 a further contract was signed with TVEL for pellets which will be incorporated into fuel assemblies at the Nuclear Fuel Complex in Hyderabad. The supply contract was renewed in January 2019.

The two small Canadian (Candu) PHWRs at **Rajasthan** nuclear power plant started up in 1972 & 1980 and are also under safeguards. Rajasthan 1 was downrated early in its life and has operated very little since 2002 due to ongoing problems. It has been shut down since 2004 as the government considers its future. It is still listed by NPCIL as operable though a parliamentary answer

in August 2012 said it “is under extended shutdown for techno-economic assessment on continuation of operations.” In March 2017 the minister said a decision on reopening Rajasthan 1 will be made following the techno-economic assessment. Rajasthan 2 was downrated in 1990. It had major refurbishment 2007-09 and has been running on imported uranium at full capacity.

The **220 MWe PHWRs** (202 MWe net) were indigenously designed and constructed by NPCIL, based on a Canadian design. The only accident to an Indian nuclear plant was due to a turbine hall fire in 1993 at Narora, which resulted in a 17-hour total station blackout. There was no core damage or radiological impact and it was rated 3 on the INES scale – a 'serious incident'.

The **Madras** (MAPS) reactors were refurbished in 2002-03 and 2004-05 and their capacity restored to 220 MWe gross (from 170). Much of the core of each reactor was replaced, and the lifespans extended to 2033/36.

Kakrapar unit 1 was fully refurbished and upgraded in 2009-10, after 16 years of operation, as was **Narora** 2, with cooling channel (calandria tube) replacement. In March 2016 unit 1 was shut down due to a coolant leak, and repairs ran through to May 2019. Kakrapar 2 was shut down in July 2015 and restarted in September 2018. There was widespread corrosion in both Kakrapar units and coolant channels were replaced.

Following the Fukushima accident in March 2011, four NPCIL taskforces evaluated the situation in India and in an interim report in July made recommendations for safety improvements of the Tarapur BWRs and each PHWR type. The report of a high-level committee appointed by the Atomic Energy Regulatory Board (AERB) was submitted at the end of August 2011, saying that the Tarapur and Madras plants needed some supplementary provisions to cope with major disasters. The two Tarapur BWRs have already been upgraded to ensure continuous cooling of the reactor during prolonged station blackouts and to provide nitrogen injection to containment structures, but further work is recommended. Madras needs enhanced flood defences in case of tsunamis higher than that in 2004. The prototype fast breeder reactor (PFR) under construction next door at Kalpakkam has defences which are already sufficiently high, following some flooding of the site in 2004.

The **Tarapur** 3&4 reactors of 540 MWe gross (490 MWe net) were developed indigenously from the 220 MWe (gross) model PHWR and were built by NPCIL. The first – Tarapur 4 – was connected to the grid in June 2005 and started commercial operation in

September. Tarapur 4's criticality came five years after pouring first concrete and seven months ahead of schedule. Its twin – unit 3 – was about a year behind it and was connected to the grid in June 2006 with commercial operation in August, five months ahead of schedule. Tarapur 3&4 cost about $1200/kW, and are competitive with imported coal.

Future indigenous PHWR reactors will be 700 MWe gross (640 MWe net). The first four are being built at Kakrapar and Rajasthan. They were due online by 2017 after 60 months' construction from first concrete to criticality, but this schedule has slipped by several years. Kakrapar 3 became the first of the four to achieve criticality in July 2020, and the unit was connected to the grid in January 2021. Up to 40% of the fuel they use will be slightly enriched uranium (SEU) – about 1.1% U-235, to achieve higher fuel burn-up – about 21,000 MWd/t instead of one-third of this. Initially this fuel will be imported as SEU.

Kudankulam 1&2: Russia's Atomstroyexport supplied the country's first large nuclear power plant, comprising two VVER-1000 (V-412) reactors, under a Russian-financed US$ 3 billion contract and 1988 Russia-India agreement with 1998 supplement. The cost was reported as Rs 17,270 crore – $2.7 billion – in 2015 but at "over Rs 22,000 crore" ($3.3 billion) by NPCIL in mid-2016, including Rs 9,000 crore escalation due to delays. A subsequent figure was Rs 20,962 crore. A long-term credit facility covered about half the cost of the plant.

The AES-92 units at Kudankulam in Tamil Nadu state have been built by NPCIL and also commissioned and operated by NPCIL under IAEA safeguards. The turbines were made by Silmash in St Petersburg and have evidently given some trouble during commissioning. Unlike other Atomstroyexport projects such as in Iran, there was only a maximum of 80 Russian supervisory staff on the project. This resulted in a more problematical than expected learning curve as Indian engineers adapted to the PWR design from Canadian-type PHWR experience. Construction started in March 2002.

Russia is supplying all the enriched fuel through the life of the plant, though India will reprocess it and keep the plutonium for civil use*. The first unit was due to start supplying power in March 2008 and go into commercial operation late in 2008, but this schedule slipped by six years. In the latter part of 2011 and into 2012 completion and fuel loading was delayed by public protests, but in March 2012 the state government approved the plant's commissioning and said it would deal with any obstruction. Unit 1 started up in mid-July

2013, was connected to the grid in October 2013 and entered commercial operation at the end of December 2014. It had reached full power in mid-year but then required turbine repairs for nearly six months. It generated only 2.8 TWh in its first year, at a cost of under Rs 4.0 per kWh (6 c/kWh). Unit 2 construction was declared complete in July 2015, it was grid-connected in August 2016, and commenced commercial operation at the start of April 2017. Each unit is 917 MWe net.

* The original agreement in 1988 specified return of used fuel to Russia, but a 1998 supplemental agreement allowed India to retain and reprocess it.

While the first core load of fuel was delivered early in 2008 there have been delays in supply of some equipment and documentation. Control system documentation was delivered late, and when reviewed by NPCIL it showed up the need for significant refining and even reworking some aspects. The design basis flood level is 5.44m, and the turbine hall floor is 8.1m above mean sea level. The 2004 tsunami was under 3m.

A small desalination plant is associated with the Kudankulam plant to produce 426 m^3/h for it using four-stage multi-vacuum compression (MVC) technology. Another reverse osmosis (RO) plant is in operation to supply local township needs.

Output from Kudankulam 1 is being supplied to India's southern grid and in 2016 divided among five states: Tamil Nadu (56%), Karnataka (22%), Kerala (13%), Andhra Pradesh (5%) and Puducherry (3%).

Kudankulam nuclear power plant, one of India's newest

Kudankulam 3&4 are being built as the first stage of phase 2 at the site and are also AES-92 units being built with Russian technical assistance "within the scope of" the 1988 agreement. Their cost is expected to be Rs 39,747 crore and the project was officially launched in October 2016 (see below under Nuclear Energy Parks). The dome of the reactor building for unit 3 was installed in December 2022. Units 3&4 are expected to be completed in 2025 and 2026 respectively.

Kaiga 3 started up in February 2007, was connected to the grid in April and went into commercial operation in May 2007. Unit 4 started up in November 2010 and was grid-connected in January 2011, but was about 30 months behind the original schedule due to a shortage of uranium. The Kaiga units are not under UN safeguards, so cannot use imported uranium. Kaiga 4 was the last of the 220 MWe PHWRs to enter service.

Rajasthan 5 started up in November 2009, using imported Russian fuel, and in December it was connected to the northern grid. RAPP 6 started up in January 2010 and was grid connected at the end of March. Both are now in commercial operation.

Under plans for the India-specific safeguards to be administered by the IAEA in relation to the civil-military separation plan, eight further reactors were to be safeguarded (beyond Tarapur 1&2, Rajasthan 1&2, and Kudankulam 1&2): Rajasthan 3&4 from 2010, Rajasthan 5&6 from 2008, Kakrapar 1&2 by 2012 and Narora 1&2 by 2014.

In mid-2008 Indian nuclear power plants were running at about half of capacity due to a chronic shortage of fuel. Average load factor for India's power reactors dipped below 60% over 2006-2010, reaching only 40% in 2008. Some easing after 2008 was due to the new Turamdih mill in Jharkhand state coming online (the mine there was already operating). Political opposition has delayed new mines in Jharkhand, Meghalaya and Telangana.

The 500 MWe **Prototype Fast Breeder Reactor** (PFBR) started construction in 2004 at Kalpakkam near Madras. It was expected to start up about the end of 2010 and produce power in 2011, but this schedule is delayed significantly. In 2014, 1750 tonnes of sodium coolant was delivered. With construction completed, in June 2015 Bhavini was "awaiting clearance from the AERB for sodium charging, fuel loading, reactor criticality and then stepping up power generation." In March 2020 the government said that commissioning would be in December 2021, but this did not proceed.

Phased core loading activities began in March 2024, with the insertion of control subassemblies and blanket subassemblies. In July of that year the regulator granted permission for the 'first approach to criticality', the next stage of commissioning which includes the loading of fuel into the reactor core and the start of low power experiments. Fuel loading began in October 2025.

In July 2025, Minister of State Jitendra Singh told India's upper house, the Rajya Sabha, that the PFBR was "at an advanced stage of commissioning" and said delays in the completion of the project were mainly due to first-of-a-kind technological issues being faced in the commissioning phase.

The approved cost is Rs 5677 crore ($850 million). It is not under IAEA safeguards. The reactor is fuelled with uranium-plutonium oxide. It has a blanket with thorium and uranium to breed fissile U-233 and plutonium respectively.

Initial FBRs will have mixed oxide fuel or carbide fuel but these will be followed by metallic fuelled ones.

In contrast to the situation in the 1990s, most PHWR reactors under construction to 2012 were on schedule (apart from fuel shortages 2007-09), and two – Tarapur 3&4 – were increased in capacity. Future PHWR units will be nominal 700 MWe (630 MWe net).

In 2005 four sites were approved for eight new reactors. Two of the sites – Kakrapar and Rajasthan – would have 700 MWe indigenous PHWR units, Kudankulam would have imported 1000 MWe VVER light water reactors alongside the two being built there by Russia, and the fourth site was greenfield for two 1000 MWe LWR units – Jaitapur (Jaithalpur) in the Ratnagiri district of Maharashtra state, on the west coast. The plan has since expanded to six 1600 MWe EPR units here.

In April 2007 the government gave approval for the first four of eight planned 700 MWe PHWR units: **Kakrapar 3&4** and **Rajasthan 7&8**, to be built by Hindustan Construction using indigenous technology. In mid-2009 construction approval was confirmed, and late in 2009 the finance for them was approved. Site works at Kakrapar were completed by August 2010. First concrete for Kakrapar 3&4 was in November 2010 and March 2011 respectively, after Atomic Energy Regulatory Board (AERB) approval. The AERB approved Rajasthan 7&8 in August 2010, and site works then began. First concrete was in July 2011. Construction was then expected to take 66 months to commercial operation. In September 2009 L&T secured an order for four steam generators

for Rajasthan 7&8, having already supplied similar ones for Kakrapar 3&4. In December 2012 L&T was awarded the $135 million contract for balance of turbine island for Rajasthan 7&8.

Their estimated cost was to be Rs 12,320 crore (Rs 123.2 billion, $2.6 billion) each pair. Both these projects were delayed apparently by the reluctance of supply chain companies to provide equipment without NPCIL giving indemnity under the 2010 Civil Liability for Nuclear Damage Act. Delays are also attributed to financial constraints. NPCIL said in July 2016 that delays in the supply of equipment including steam generators from Indian sources plus the nuclear liability issue have put the projects behind schedule, and the Minister of Atomic Energy said that Kakrapar 3&4 were only 75.5% completed and Rajasthan 7&8 were only 61.5% completed then. In March 2020 the government said that Kakrapar 3&4 and Rajasthan 7&8 were expected to be commissioned in October 2020, September 2021, March 2022 and March 2023, respectively. In July 2020, Kakrapur 3 achieved first criticality, and was connected to the grid in January 2021. Kakrapar 4 was connected to the grid in February 2024. Rajasthan 7 was connected to the grid in March 2025. Rajasthan 8 is expected to be connected to the grid in 2026 respectively.

Construction costs of reactors as reported by the AEC are about $1200 per kilowatt for Tarapur 3&4 (540 MWe), $1300/kW for Kaiga 3&4 (220 MWe) and expected $1700/kW for the 700 MWe PHWRs with 60-year life expectancy.

In April 2015 the government gave in principle approval for new nuclear plants at ten sites in nine states. Those for indigenous PHWRs are: Gorakhpur in Haryana's Fatehabad; Chutka and Bhimpur in Madhya Pradesh; Kaiga in Karnataka; and Mahi Banswara in Rajasthan. Those for plants with foreign cooperation are: Kudankulam in Tamil Nadu (VVER); Jaitapur in Maharashtra (EPR); Chhaya Mithi Virdhi in Gujarat (AP1000); Kovvada in Andhra Pradesh (originally ESBWR) and Haripur in West Bengal (VVER), though this location had been in doubt. In addition, two 600 MWe fast breeder reactors are proposed at Kalpakkam. (All these are in line with the 12th five-year plan for 2012-17, with the addition of Bhimpur, which in 2017 was omitted in favour of two extra units at Mahi Banswara.) In mid-2016 the Kovvada site was allocated for AP1000 units instead of Mithi Virdhi, and the ESBWR prospects receded.

New phase of nuclear industry

developments

Following the Nuclear Suppliers Group agreement which was achieved in September 2008, the scope for sourcing both reactors and fuel from suppliers in other countries opened up. Civil nuclear cooperation agreements have been signed with the USA, Russia, France, UK, South Korea, Czech Republic and Canada, as well as Australia, Argentina, Kazakhstan, Mongolia and Namibia. A further nuclear cooperation agreement was signed with the UK in November 2015, with “a comprehensive package” of collaboration on energy and climate change matters involving £3.2 billion ($4.9 billion) in programs and initiatives related to energy security and energy access. However, there was no civil nuclear cooperation agreement with Japan, which loomed as a limiting factor for some technology provision involving GE Hitachi and Westinghouse. Eventually a preliminary agreement was signed in December 2015, and after six years of negotiations a full nuclear cooperation agreement was signed in November 2016. It will allow India to import Japanese nuclear technology, and secures Japan’s support for India to join the international Nuclear Suppliers Group (NSG). A civil nuclear cooperation agreement was signed with Japan in 2016 and passed by Japan’s parliament in June 2017.

On the basis of the 2010 cooperation agreement with Canada, in April 2013 a bilateral safeguards agreement was signed between the Department of Atomic Energy (DAE) and the Canadian Nuclear Safety Commission (CNSC), allowing trade in nuclear materials and technology for facilities which are under IAEA safeguards. A similar bilateral safeguards agreement with Australia was signed in 2014 and finalized in November 2015. Both apply essentially to uranium supply.

The initial two Russian PWR types at the Kudankulam site were apart from India's three-stage plan for nuclear power and were simply to increase generating capacity more rapidly. Now there are plans for eight 1000 MWe units at that site, and in January 2007 a memorandum of understanding was signed for Russia to build the next four there, as well as others elsewhere in India. A further such agreement was signed in December 2010, and Rosatom announced that it expected to build no less than 18 reactors in India. Then in December 2014 another high-level nuclear cooperation agreement was signed with a view to Russia building 20 more reactors plus cooperation in building Russian-designed nuclear power plants in third countries, in uranium mining, production of nuclear fuel, and waste management. India was also to confirm a second location for a Russian plant – Haripur in West

Bengal being in some doubt. Most of the new units are expected to be the larger 1200 MWe AES-2006 designs. Russia was earlier reported to have offered a 30% discount on the $2 billion price tag for each of the phase 2 Kudankulam reactors. This was based on plans to start serial production of reactors for the Indian nuclear industry, with much of the equipment and components proposed to be manufactured in India, thereby bringing down costs. However, at the end of 2015 the approved cost of Kudankulam units 3&4 was Rs 39,747 crore ($5.96 billion), according to the Minister for Atomic Energy, more than twice the costs of units 1&2, due to liability issues.

Between 2010 and 2020, further nuclear plant construction was expected to take total gross capacity to 21,180 MWe. The planned increment included many of those set out in the Table below ('planned') plus the initial 300 MWe advanced heavy water reactor (AHWR).

Looking beyond the Russian light water reactors, NPCIL had meetings and technical discussions with three major reactor suppliers – Areva, GE Hitachi, and Westinghouse Electric Corporation – for the supply of reactors for these projects and for new units at Kaiga. These resulted in more formal agreements with each reactor supplier early in 2009, as described in the Nuclear Energy Parks subsection below. The benchmark capital cost sanctioned by DAE for imported units was quoted at $1600 per kilowatt. An important aspect of all these agreements is that, as with Kudankulam, India will reprocess the used fuel to recover plutonium for its indigenous three-stage civil program, using a purpose-built and safeguarded integrated nuclear recycle plant. However, all three agreements beyond that with Russia are stalled due to liability concerns.

In late 2008 NPCIL announced that as part of the Eleventh Five Year Plan (2007-12), it would start site work for 12 reactors including the rest of the eight 700 MWe PHWRs, three or four fast breeder reactors and one 300 MWe advanced heavy water reactor (AHWR) in 2009. NPCIL said that "India is now focusing on capacity addition through indigenization" with progressively higher local content for imported designs, up to 80%. Looking further ahead its augmentation plan included construction of 25-30 light water reactors of at least 1000 MWe by 2030. In the event only four 700 MWe PHWR units started construction over 2007-12.

Early in 2012 NPCIL projections had the following additions to the 10.08 GWe anticipated in 2017 as 'possible': 4.2 GWe PHWR, 7.0

GWe PHWR (based on recycled U), 40 GWe LWR, 2.0 GWe FBR. These projections also have not materialized.

In June 2012 NPCIL announced four new sites for twin PHWR units: at Gorakhpur/Kumbariya near Fatehabad district in Haryana, at Banswara in Rajasthan, at Chutka in Mandla district and at Bhimpur also in Madhya Pradesh. Initially these would add 2800 MWe, followed by a further 2800. Site work has started at Gorakhpur with Haryana state government support.

In mid-2015 NPCIL confirmed plans for Kaiga 5&6 as 700 MWe PHWR units, costing about Rs 6000 crore (Rs 60 billion). In September 2019 India's Ministry of Environment, Forest and Climate Change (MoEF) approved NPCIL's plans. The turbine islands and steam generators for Kaiga 5&6 were ordered from BHEL in July and August 2021, and in May 2022 excavation works began. In April 2025 NPCIL awarded an engineering, procurement and construction (EPC) contract for the two units to Megha Engineering & Infrastructure.

NPCIL is also planning to build an indigenous 900 MWe PWR, the Indian Pressurized Water Reactor (IPWR), designed by Bhabha Atomic Research Centre (BARC) in connection with its work on submarine reactors. A site for the first plant is being sought, a uranium enrichment plant is planned, the reactor pressure vessel forging will be carried out at Larsen & Toubro (L&T) and NPCIL's joint venture heavy forge unit at Hazira in Gujarat, and the turbine will come from Bharat Heavy Electricals Limited (BHEL).

Meanwhile, NPCIL is offering both 220 and 540 MWe PHWRs for export, in markets requiring small- to medium-sized reactors.

Power reactors planned

Reactor	State	Type	MWe gross
Chutka 1&2	Madhya Pradesh (Mandla)	PHWR x 2	700 x 2
Gorakhpur 1&2	Haryana (Fatehabad district)	PHWR x 2	700 x 2
Gorakhpur 3&4	Haryana (Fatehabad district)	PHWR x 2	700 x 2
Kaiga 5&6	Karnataka	PHWR x 2	700 x 2
Kalpakkam 1&2	Tamil Nadu	FBR x 2	500 x 2
Mahi Banswara 1&2	Rajasthan	PHWR x 2	700 x 2
Mahi Banswara 3&4	Rajasthan	PHWR x 2	700 x 2
Tarapur BSMR	Maharashtra	PHWR	220

Reactor	State	Type	MWe gross
Tarapur SMR-55	Maharashtra	PWR?	55
Subtotal planned		14 units	9675 MWe

Power reactors proposed

Reactor	State	Type	MWe gross (each)	Project control
Tarapur?	Maharashtra?	AHWR	300	NPCIL
"Haripur 1&2" another site	West Bengal (but likely to be relocated, maybe to Kavali in Andhra Pradesh)	AES-2006	1200	NPCIL
Kudankulam 7&8	Tamil Nadu	AES-2006	1200	NPCIL
"Kudankulam 9-12"	Andhra Pradesh	AES-2006	1200	NPCIL
Bhimpur 1-4	Madhya Pradesh	PHWR	700	NPCIL
Chutka 3&4	Madhya Pradesh	PHWR	700	BHEL-NPCIL-GE?
Rajouli, Nawada 1&2	Bihar	PHWR	700	NPCIL
?		PWR x 2	1000	NPCIL/NTPC
Jaitapur 1&2	Ratnagiri, Maharashtra	PWR – EPR	1700	NPCIL
Jaitapur 3&4	Ratnagiri, Maharashtra	PWR – EPR	1700	NPCIL
?	?	FBR x 4	500	Bhavini
Jaitapur 5&6	Ratnagiri, Maharashtra	PWR – EPR	1700	NPCIL
Markandi (Pati Sonapur)	Orissa	PWR 6000 MWe		NPCIL
Kovvada 1&2	Srikakulam, Andhra Pradesh	AP1000	1250	NPCIL
Kovvada 3&4	Srikakulam, Andhra Pradesh	AP1000	1250	NPCIL
Earlier: "Kovvada 1-6"	Originally Srikakulam, Andhra Pradesh	Originally ESBWR	1600	NPCIL

Reactor	State	Type	MWe gross (each)	Project control
Nizampatnam 1-6	Guntur, Andhra Pradesh	6x?	1200	NPCIL
"Haripur 3-6" another site	West Bengal, Orissa or Kavali in Andhra Pradesh	AES-2006?	1200	NPCIL
Pulivendula	Kadapa, Andhra Pradesh	PWR? PHWR?	1000? 700?	NPCIL 51%, AP Genco 49%
Kovvada 5&6	Srikakulam, Andhra Pradesh	AP1000	1250	NPCIL
Chhaya-Mithi Virdi 1-6	Bhavnagar, Gujarat	AP1000	1250	NPCIL

List of proposed units includes all known proposals made to date for completeness. Not all units listed as proposed in the above table are likely to be built. The government of India in March 2018 identified 28 units and 32 GWe of capacity as approved "in principle".[d] The government in 2025 set a target for nuclear capacity to reach 100 GWe by 2047, in the centenary year of the country's independence. For World Nuclear Association's reactor table, these figures are listed as 'proposed', with the exception of Kalpakkam 1&2, which have been promoted to 'planned'.

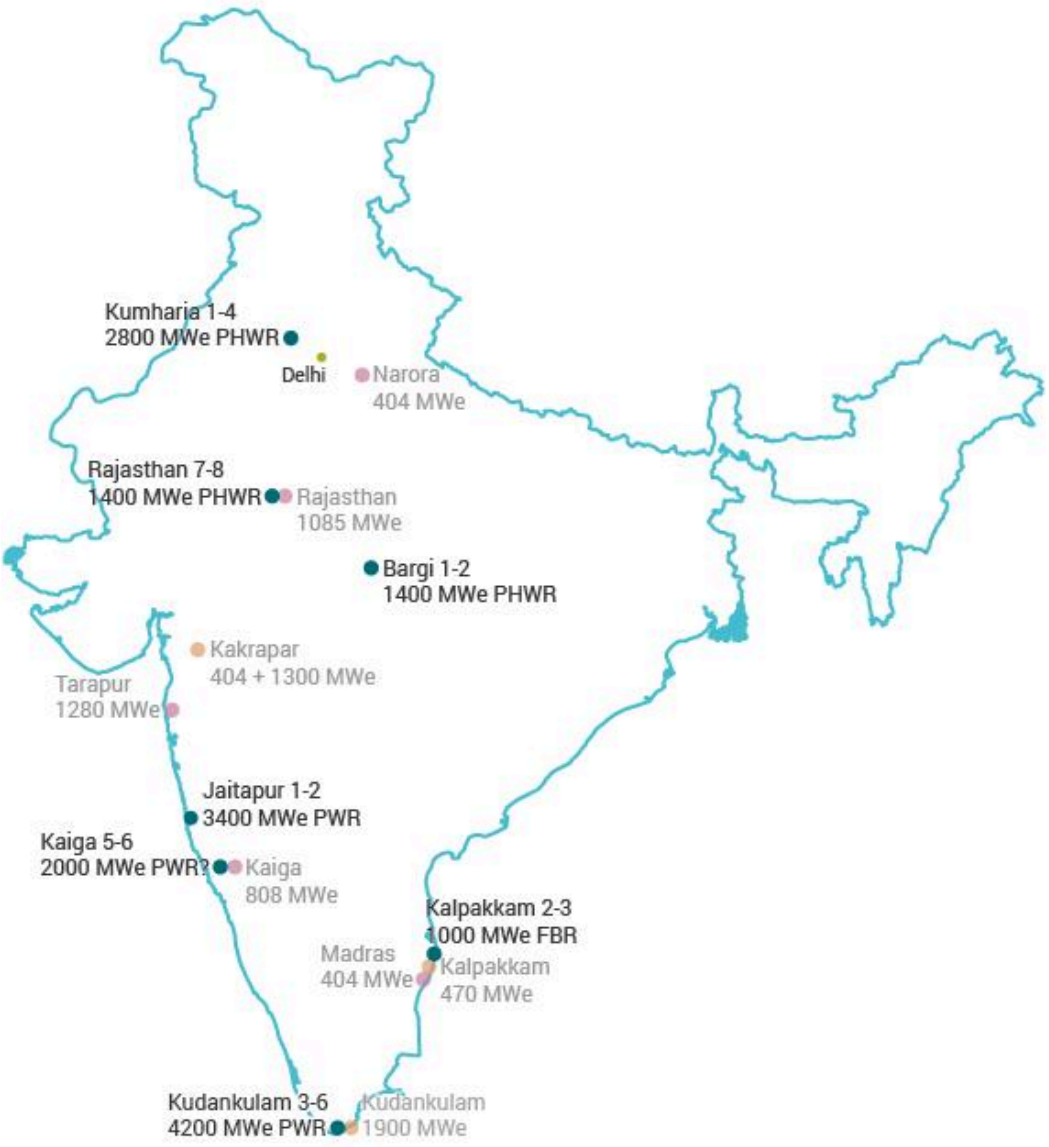
Nuclear Energy Parks

In line with past practice such as at the eight-unit Rajasthan nuclear plant, NPCIL intends to set up five further 'Nuclear Energy Parks', each with a capacity for up to eight new-generation reactors of 1000 MWe, six reactors of 1600 MWe or simply 10,000 MWe at a single location. By 2032, 40-45 GWe would be provided from these five. NPCIL was hoping to be able to start work by 2012 on at least four new reactors at all four sites designated for imported plants, but this did not happen. In mid-2015 it was reported that an additional site could be assigned for a Japanese multi-unit plant. However, apart from the Russian projects under intergovernmental agreement, no overseas reactor vendor has been ready to proceed under India's unique liability arrangements.

Original plans were for widespread deployment of new nuclear capacity, but due to protests in Gujarat, Tamil Nadu, West Bengal and Maharashtra, some of the plans have relocated proposed developments to Andhra Pradesh. That state may now host six

Russian reactors moved from Haripur in West Bengal to Kavali in Nellore district, six Westinghouse AP1000 moved from Mithi Virdi in Gujarat to Kovvada in Srikakulam district, as well as the original six GE Hitachi ESBWR units if they are ever approved.

Planned nuclear power plants in India



The five new energy parks include two that are proceeding and one that shows some promise:

Kudankulam (KKNPP) **phases 2, 3&4** in Tamil Nadu: three more pairs of Russian AES-92 units with VVER-1000 reactors, making eight, with 9200 MWe were envisaged, but it now appears that after six 1000 MWe reactors, units 7-8 will be the larger AES-2006 design with VVER-1200 reactors. Agreements intended for mid-2010 were delayed on account of supplier liability questions, with India wanting the units to come under its 2010 vendor liability law. In July 2012 coastal regulation zone clearance was obtained for units 3-6 of 1000 MWe each from the Ministry of Environment &

Forests, mainly related to seawater cooling. Environmental approval for units 3-6 had been obtained earlier.

Phase 2: In July 2012 Russia agreed to $3.5 billion in export financing for units 3&4, to cover 85% of their cost. A further credit line of $800 million is available to cover fuel supplies. The credit lines carry interest at 4% pa and would be repayable over 14 years and 4 years respectively, from one year after the start of power generation. The Indian government said it expected to take up the credit offers to the value of $3.06 billion, about 53% of the then estimated $5.78 billion total project cost.

In March 2013 cabinet approved construction of units 3&4, and some site preparation began. In April 2014 NPCIL signed a Rs 33,000 crore ($ 5.47 billion) agreement with Rosatom for units 3&4, having apparently resolved the liability question (but see section below). In May a general framework agreement to build the plants was signed, and in December contracts with Rosatom were signed. Rosatom said that the general contractor is Atomstroyexport, and the general designer is Atomenergoproekt. In September 2015 Rosatom contracted Atomenergomash for the complete supply of major components for the two reactors, to be delivered to the plant over 2016-2018. OMZ-Spetsstal completed the pressure vessel forgings for unit 3 in June 2016 and sent them to OMZ Izhorskiye Zavody for fabrication, with internals. AEM-Technology's Volgodonsk branch (Atommash) will supply the reactor pressure vessel for unit 4 and steam generators for both units, manufacture of which began in November 2016, for shipment in 2018. Some ancillary equipment is from ZiO-Podolsk. Larsen & Toubro is contracted for civil works.

In August 2015 the government said that all issues had been resolved to enable construction of units 3&4 to start. Excavation started in February 2016, the AERB construction permit was issued with construction start a week later in June 2017. A 72-month construction period was expected, under NIAEP-ASE supervision. The project was officially launched in October 2016. The approved project cost is Rs 39,747 crore ($6.25 billion), about double per MW that of established PHWR plants, but using the ruble as currency peg. Generation cost is expected to be about Rs 3.9/kWh (5.8 cents/kWh), competitive with coal. In March 2020 the government said that the units would be commissioned in March 2023 and November 2023.

Phase 3: In October 2016 a general agreement was signed for units 5&6, along with finalizing a credit protocol. A framework agreement was signed in June 2017, including an

intergovernmental credit protocol, so that the project entered what Rosatom called the “practical implementation phase”. At the end of July ASE Group and NPCIL signed contracts covering the design and supply of the main plant components for units 5&6. In February 2021 NPCIL contracted Larsen & Toubro for civil works over 64 months. First concrete for unit 5 was in June 2021 and unit 6 in December 2021.

Phase 4: Plans are for units 7&8 to be AES-2006 units with VVER-1200 reactors.

In 2015, due to the nuclear liability law constraints on other foreign reactor vendors, four more Russian units were agreed. However, though broadly discussed as 'Kudankulam', these will be in Andhra Pradesh, possibly at Kavali in Nellore district, and will likely be AES-2006 power plants with VVER-1200 reactors. See Haripur, below.

Gorakhpur Haryana Anu Vidyut Pariyojana (GHAVP) in the Fatehabad district of Haryana is a project with four indigenous 700 MWe PHWR units in two phases, and the AEC has approved the state's proposal for the 2800 MWe plant. Kakrapar 3&4 and Rajasthan 7&8 are the reference design. The inland northern state of Haryana is one of the country's most industrialized and has a demand of 8900 MWe, but currently generates less than 2000 MWe and imports 4000 MWe. The Gorakhpur plant may be paid for by the state government or Haryana Power Generation Corp.

NPCIL is undertaking site infrastructure works near the villages of Kumharia and Gorakhpur, and the official groundbreaking was in January 2014. A final environmental assessment for the project was approved in December 2013, and government approval for Gorakhpur phase 1 was in February 2014. The AERB granted a siting licence in July 2015. Construction was initially due to begin in June 2015, with the first unit online in 2021. The Minister of State for Atomic Energy and Space announced in March 2016 that the first unit would be online in six years, and electricity would be supplied to consumers at the rate of Rs 6.5 ($0.10) per unit. The cost of the first two units was then put at Rs 210 billion ($3.4 billion). In July 2016 NPCIL was seeking bids for domestic supply of equipment, noting that this would be the first plant subject to the 2010 civil liability legislation. NPCIL commenced early site works in 2018, and in November 2020 the AERB approved the pouring of first nuclear safety-related concrete. The turbine islands and steam generators for Gorakhpur 1-4 were ordered from BHEL in July and August 2021.

In June 2025, the Ministry of Power said first concrete for unit 1 was targeted for October 2025, with first criticality by June 2030 and commercial operation by March 2031. Unit 2 is expected to follow about six months later. In August 2025, Minister of State Jitendra Singh told the Lok Sabha that remediation of localized weak zones in soil strata discovered during site excavations had led to the start of nuclear island construction being delayed, but repeated that completion of Gorakhpur 1&2 was expected by 2031-32.

Three projects are delayed indefinitely, though one may proceed after site reallocation:

Kovvada in Andhra Pradesh's northern coastal Srikakulam district was originally intended to host six GE Hitachi ESBWR units but was subsequently designated for six Westinghouse AP1000 units.

GE Hitachi said in June 2012 that it was undertaking a preliminary environmental assessment and preparing an early works agreement with NPCIL to set terms for obtaining approval from the government for the project. In February 2014 NPCIL said it hoped to commence construction of the first 1594 MWe reactor early in 2015. However, with no change to the 2010 Civil Liability for Nuclear Damage Act, GEH in September 2015 said it would not proceed with any investment in India until the country's liability regime was brought into line with the rest of the world. In June 2016 DAE said that it would not support building any reactor design that did not have a reference plant, which ruled out the ESBWR for the time being. This coincided with NPCIL allocating the Kovvada site to Westinghouse for six AP1000 reactors.

In July 2016 the Atomic Energy Minister said that a land survey had been completed and acquisition of 840 ha was proposed. In September 2016 NPCIL submitted an environmental assessment to the environment ministry, requesting clearance to proceed with building the AP1000 plant. A draft social impact study has been completed for the state government. The Indian and US governments called for continued engagement between Westinghouse and NPCIL with the intention to finalize contractual arrangements for the six Kovvada units by June 2017. In February 2025 the Indian prime minister and American president reaffirmed their intention to advance the project.

Jaitapur (JNPP) in Maharashtra's Ratnagiri district: following a February 2009 general agreement with Areva (now Framatome) to build six EPR reactors, a €7 billion framework agreement was signed in December 2010 for the first two, with Alstom turbine-generators, along with 25 years supply of fuel.

Environmental approval has been given for these, with coastal zone clearances. Land acquisition was completed in 2018. The site could host six units, providing up to 9600 MWe. Areva had hoped to obtain export credit financing and sign a contract by the end of 2012, to put the first two units online in 2020 and 2021. In 2013 negotiations continued and the government said it expected the cost of the first two units to be 120,000 crore ($20 billion). France has agreed to a 25-year loan for the project at 4.8%. In April 2015 Areva signed a pre-engineering agreement contract with NPCIL in preparation for licensing the EPR design. In May 2015 Areva said that construction might begin in two years, with 50% local content in the first units. However, timing would be dependent on the resolution of nuclear liability questions.

In January 2016 the prime minister and the French president announced that they "encouraged their industrial companies to conclude techno-commercial negotiations by the end of 2016" on Jaitapur. They called for "due consideration to cost viability of the project, economical financing from the French side, collaboration on transfer of technology and cost-effective localization of manufacturing in India for large and critical components. Their shared aim is to start the implementation of the project in early 2017." France acknowledges the need for India to have a "lifetime guarantee of fuel supply, and renewed its commitment to reliable, uninterrupted and continued access to nuclear fuel supply throughout the entire lifetime of the plants." In July 2016 EdF submitted a fresh proposal to NPCIL and the Ministry of External Affairs for six EPR units, but seeking guarantee of “the same level of protection” in relation to liability that is available at the international level, and citing the Vienna convention on liability. In mid-2017 EdF said it expected to sign a general framework agreement with NPCIL by the end of the year.

In March 2018, a so-called Industrial Way Forward Agreement was signed by EDF and NPCIL, setting out the industrial framework and planned timetable for the six EPR reactors at Jaitapur. Under its terms, EDF will act as supplier of the EPR technology, and will undertake all engineering studies and component procurement activities for the first two of the six reactors. Responsibility for some purchasing activities and studies for the other four units may be assigned to local companies, reaching potential Indian localization of 60% for the last two of the six reactors. As owner and future operator of the Jaitapur plant, NPCIL will be responsible for obtaining all authorizations and certifications required in India, and for constructing all six reactors and site infrastructures. Upon signing of the agreement, a goal of commencing works at Jaitapur

"around the end of 2018" was reiterated. In June 2018, GE and EDF signed a strategic cooperation agreement, described by the two companies as "an important step in implementing the Industrial Way Forward Agreement." GE Power will design the conventional island at Jaitapur and supply its main components. EDF will be responsible for engineering integration across the entire project.

In May 2022 a communique after a meeting of the French president and Indian prime minister reaffirmed their commitment to the success of the Jaitapur project.

Chhaya-Mithi Virdi in Gujarat's Bhavnagar district was intended to host up to six Westinghouse AP1000 units built in three stages on the coast, and it may yet do so.

NPCIL commenced site works in 2012, and a preliminary environmental assessment for the whole project was completed in January 2013. State and local government and coastal zone clearances were obtained. A preliminary commercial contract between NPCIL and Westinghouse was signed in September 2013 along with an agreement to carry out a two-year preliminary safety analysis for the project. NPCIL said that it “must lay emphasis on strong public acceptance outreach and project planning." In October 2014 the Ministry of Environment & Forests asked NPCIL for further assessment of environmental and land acquisition matters in its environment impact assessment (EIA). NPCIL was then in the process of obtaining site clearance from the Atomic Energy Regulatory Board (AERB). However, the land acquisition process was held up pending passage of a new federal Land Acquisition Act, which was delayed in the upper house. Then in May 2016 NPCIL changed the initial Westinghouse AP1000 site to Kovvada in the northern coastal Srikakulam district of Andhra Pradesh, and in June the government announced that the final contract was to be completed in June 2017. The first stage of two units was originally due online in 2019-20, the others to 2024. Westinghouse still expects to build six AP1000 units at Chhaya-Mithi Virdi however, after those at Kovvada**.**

In addition to the original five energy parks, there are four proposals and the first two are considered 'planned':

Chutka (CNPP) in inland Madhya Pradesh is also designated for two indigenous 700 MWe PHWR units. NPCIL has initiated pre-project activities here. The EIA report was released in March 2013 and a public hearing at Chutka was in February 2014. The expected cost for two units was then set at Rs 16,550 crore ($2.48

billion). Construction start was planned for 2015, now likely in the early 2020s.

Mahi Banswara in Rajasthan is a new site for 700 MWe PHWRs. Two units were in the 2012 XII Plan, and the government announced in principle approval in August 2016 for the construction of four units. This was confirmed in July 2017. In May 2025 India's regulator gave siting consent for the four-unit plant.

At **Markandi** (Pati Sonapur) in Orissa there are plans for up to 6000 MWe of PWR capacity. Major industrial developments are planned in that area and Orissa was the first Indian state to privatize electricity generation and transmission. State demand was expected to reach 20 billion kWh/yr by 2010. However, these plans may have merged with others.

Bhimpur in Madhya Pradesh has in-principle government approval for two 700 MWe PHWRs, according to the DAE annual report 2013-14, but by 2017 plans these had been transferred to Mahi Banswara.

The AEC has also mentioned possible new nuclear power plants in Bihar and Jharkhand.

Haripur in West Bengal: was to host four or six further Russian **VVER-1200** units. NPCIL has initiated pre-project activities here, and groundbreaking was planned for 2012. However, strong local opposition led the West Bengal government to reject the proposal in August 2011, and change of site to Orissa state was suggested. In 2015 Andhra Pradesh and Karnataka had expressed interest in hosting further Russian plants. Certainly Rosatom expects to build six further Russian VVER reactors at a new site, not yet identified, and hopes to build up to 14 more after that. Haripur is abandoned. In October 2016 the government and Rosatom said that allocation of a new site was likely soon. Kavali in Andhra Pradesh appears most likely.

In 2014 the Chinese president initiated discussions with his Indian counterpart about building nuclear power plants, raising he possibility that China could compete with France, Russia, Japan and the USA.

Fast neutron reactors

In the longer term, the AEC envisages its fast reactor programme being 30 to 40 times bigger than the PHWR programme, and initially at least, largely in the military sphere until its "synchronized

working" with the reprocessing plant is proven on an 18- to 24-month cycle. This would be linked with up to 40,000 MWe of light water reactor capacity, the used fuel feeding ten times that fast breeder capacity, thus "deriving much larger benefit out of the external acquisition in terms of light water reactors and their associated fuel". This 40 GWe of imported LWR capacity multiplied to 400 GWe via FBR would complement 200-250 GWe based on the indigenous three-stage programme of PHWR-FBR-AHWR (see section on Thorium fuel cycle below). Thus AEC is "talking about 500 to 600 GWe nuclear over the next 50 years or so" in India, plus export opportunities.

In 2002 the regulatory authority issued approval to start construction of a 500 MWe prototype fast breeder reactor (PFBR) at Kalpakkam and this has been built by BHAVINI (Bharatiya Nabhikiya Vidyut Nigam Ltd), a government enterprise set up under the DAE to focus on FBRs. It was expected to start up in September 2014, fuelled with MOX (mixed uranium-plutonium oxide, the 30% of reactor-grade Pu being from its existing PHWRs) made at Tarapur by BARC, as hexagonal fuel asemblies. It has a blanket with uranium and thorium to breed fissile plutonium and U-233 respectively, taking the thorium programme to stage two, and setting the scene for eventual full utilization of the country's abundant thorium to fuel reactors. It is a sodium-cooled pool-type reactor having two primary and two secondary loops, with four steam generators per loop. It is designed for a 40-year operating lifetime at 75% load factor. Two more such 500 MWe fast reactors have been announced for construction at Kalpakkam, but slightly redesigned by the Indira Gandhi Centre to reduce capital cost. Then four more are planned at another site.

Initial FBRs will have mixed oxide fuel or carbide fuel, but these will be followed by metallic fuelled ones to enable shorter doubling time. One of the last of the above six, or possibly the fourth one overall, is to have the flexibility to convert from MOX to metallic fuel (*i.e.* a dual fuel unit), and it was planned to convert the small FBTR to metallic fuel about 2013 (see R&D section below). With metal fuel, a 500 MWe unit is expected to produce 2 tonnes of reactor-grade plutonium in 8-10 years. The reactor is not under international safeguards.

Following these will be a 1000 MWe fast reactor using metallic fuel. A fuel fabrication plant and a reprocessing plant for metal fuels are planned for Kalpakkam in 2024, known as the Fast Reactor Fuel Cycle Facility (FRFCF). See below.

A December 2010 scientific and technical cooperation agreement between the AEC and Rosatom is focused on "joint development of a new generation of fast reactors".

NPCIL collaboration with other state entities

In February 2016 the government amended the Atomic Energy Act to allow NPCIL to form joint venture companies with other public sector undertakings (PSUs) for involvement in nuclear power generation and possibly other aspects of the fuel cycle. The legislative change does not extend to private sector companies, and nor does it allow direct foreign investment in nuclear power, apart from the supply chain. This is expected to help NPCIL secure funding for new projects.

Three joint venture companies involving major PSUs in the energy area have been incorporated and are able to come into effect under the revised legislation: Anushakti Vidhyut Nigam Ltd (NPCIL and NTPC), NPCIL-Indian Oil Nuclear Energy Corporation Ltd, and NPCIL-Nalco Power Company Ltd. NPCIL itself is reported to have about Rs 12,000 crore ($1.8 billion) of investible surplus; the three other PSUs – NTPC, IOC and Nalco – are able to contribute about Rs 10,000 crore ($1.5 billion) each to new nuclear projects in which they will have 49% equity. Some of this will be in hard currency, which may be why the focus has shifted from indigenous PHWR plants (which can be paid for in local currency) to new plants with imported LWR technology.

NTPC plans

India's largest power company, National Thermal Power Corporation (NTPC) in 2007 had proposed building a 2000 MWe nuclear power plant to be in operation by 2017. It would be the utility's first nuclear plant and also the first conventional nuclear plant not built by the 89.5% government-owned NPCIL. This proposal took the form of a joint venture in 2011 with NPCIL holding 51%, and possibly extending to multiple projects utilizing local and imported technology, but pending amendment to the Atomic Energy Act. One of the sites earmarked for a pair of 700 MWe PHWR units in Haryana or Madhya Pradesh was considered prospective for the joint venture. NTPC said it aimed by 2014 to have demonstrated progress in "setting up nuclear power generation capacity", and that the initial "planned nuclear portfolio

of 2000 MWe by 2017" could be greater. However in 2012 it indicated a downgrading of its nuclear plans.

NTPC planned to increase its total installed capacity to 70 GWe by 2017 and 128 GWe by 2032, from 47 GWe (74% coal) in 2016. In 2008 it also formed joint ventures in heavy engineering, with BHEL and Bharat Forge. The former is to explore, secure and execute EPC contracts for power plants and other infrastructure projects in India and abroad, as well as manufacturing and supplying equipment for them. With the 2011 JV with NPCIL, this was reported as also selling India's largely indigenous 220 MWe PHWR reactor units abroad, possibly in contra deals involving uranium supply from countries such as Namibia and Mongolia.

Nalco plans

The 87% state-owned National Aluminium Company (Nalco) signed an agreement with NPCIL with the intention of building a 1400 MWe nuclear power plant on the east coast, in Orissa's Ganjam district. A more specific agreement was signed in November 2011 to set up a joint venture with NPCIL – NPCIL Nalco Power Co Ltd – giving it 26% equity in Kakrapar 3&4 (total 1300 MWe net) under construction in Gujarat on the west coast for Rs 1700 crore ($285 million). The total project size is Rs 12,000 crore with the total debt requirement at Rs 7,000 crore. Nalco sought government permission to increase this share to 49%, pending amendment to the Atomic Energy Act. It was also seeking to buy uranium assets in Africa. In May 2016 it was reported to have pulled out of the JV with NPCIL, but at the end of the year the JV continued.

Nalco already has its own 1200 MWe coal-fired power plant in Orissa state at Angul, to serve its refinery and its Angul smelter of 345,000 tpa, being expanded to 460,000 tpa (requiring about 1 GWe of constant supply). It has set up wind farms in Andhra Pradesh (50.4 MWe) and Rajasthan (47.6 MWe).

IOC and ONGC plans

India's national oil company, Indian Oil Corporation Ltd (**IOC**), in November 2009 joined with NPCIL in an agreement "for partnership in setting up nuclear power plants in India," anticipating an amendment to the Atomic Energy Act in 2016. The initial plant envisaged was to be at least 1000 MWe, and NPCIL

would be the operator and at least 51% owner. In November 2010 IOC agreed to take a 26% stake in Rajasthan 7&8 (2x700 MWe) as a joint venture, with the option to increase this to 49%. The estimated project cost is Rs 12,320 crore (123 billion rupees, $2.1 billion), and the 26% stake represented only 2% of IOC's capital budget in the 11th plan to 2012. The formal joint venture agreement was signed in January 2011.

The cash-rich Oil and Natural Gas Corporation (**ONGC**), which (upstream of IOC) provides some 80% of the country's crude oil and natural gas and is 84% government-owned, submitted a plan to collaborate with NPCIL in October 2017. The plan follows the government's amendment to the Atomic Energy Act 1962 to enable NPCIL to form joint ventures with public companies. It is expected that ONGC will become a minority partner with NPCIL on a number of present or planned 700 MWe PHWR projects.

Other plans and proposals

Indian Railways, with power requirement of 3000 MWe now and rising to 5000 MWe about 2022, also approached NPCIL to set up a joint venture to build two 500 MWe PHWR nuclear plants on railway land or existing nuclear sites for its own power requirements. The Railways already has a joint venture with NTPC – Bhartiya Rail Bijlee Company – to build a 1000 MWe coal-fired power plant at Nabi Nagar in Aurangabad district of Bihar, with the 250 MWe units coming on line 2014-15. The Railways also plans to set up another 2 x 660 MWe supercritical thermal power plant at Adra in Purulia district of West Bengal for traction supply at economical tariff. Some 23,500 km of its 65,000 km lines are electrified, and it spends 8000 crore ($1.34 billion) per year on power, at INR 5.4/kWh which it expects to reduce to INR 4.0/kWh (9 cents to 6.6 c).

In March 2017 NPCIL said it planned a joint venture with Indian Railways to set up nuclear power projects, but this is apparently not proceeding.

The Steel Authority of India Ltd (**SAIL**) and NPCIL were discussing a joint venture to build a 700 MWe PHWR plant. The site would be chosen by NPCIL, in Gujarat or elsewhere in western India.

In anticipation of the Atomic Energy Act amendment in 2016, Reliance Power Ltd, GVK Power & Infrastructure Ltd and GMR Energy Ltd were reported to be in discussion with overseas nuclear

vendors including Areva, GE Hitachi, Westinghouse and Atomstroyexport.

In September 2009 the AEC announced a version of its planned Advanced Heavy Water Reactor (the AHWR-300 LEU) designed for export.

In August and September 2009 the AEC reaffirmed its commitment to the thorium fuel cycle, particularly thorium-based FBRs, to make the country a technological leader. However, little has happened on this front since then.

Overseas reactor vendors

As described above, there have been a succession of agreements with Russia's Atomstroyexport to build further VVER reactors. In March 2010 a 'roadmap' for building six more reactors at Kudankulam by 2017 and four more at Haripur after 2017 was agreed, bringing the total to 12. Associate company Atomenergomash (AEM) set up an office in India with a view to bidding for future work there and in Vietnam, and finalizing a partnership with an Indian heavy manufacturer, either L&T (see below) or another. A Russian fuel fabrication plant is also under consideration.

In February 2009 Areva (now Framatome) signed a memorandum of understanding with NPCIL to build two, and later four more, EPR units at Jaitapur, and a formal contract was expected but this did not proceed. This followed the government signing a nuclear cooperation agreement with France in September 2008.

In March 2009 GE Hitachi Nuclear Energy signed agreements with NPCIL and Bharat Heavy Electricals Ltd (BHEL) to begin planning to build a multi-unit power plant using 1350 MWe Advanced Boiling Water Reactors (ABWR). In May 2009 L&T was brought into the picture. In April 2010 it was announced that the BHEL-NPCIL joint venture was still in discussion with an unnamed technology partner to build a 1400 MWe nuclear plant at Chutka in Madhya Pradesh state, with Madhya Pradesh Power Generating Company Limited (MPPGCL) the nodal agency to facilitate the execution of the project.

In May 2009 Westinghouse signed a memorandum of understanding with NPCIL regarding deployment of its AP1000 reactors, using local components (probably from L&T).

After a break of three decades, Atomic Energy of Canada Ltd (AECL) was keen to resume technical cooperation, especially in

relation to servicing India's PHWRs (though this would now be undertaken by Candu Energy), and there were preliminary discussions regarding the sale of an ACR-1000.

In August 2009 NPCIL signed agreements with Korea Electric Power Co (KEPCO) to study the prospects for building Korean APR-1400 reactors in India. This could proceed following bilateral nuclear cooperation agreements signed in October 2010 and July 2011.

The LWRs to be set up by these foreign companies are reported to have a lifetime guarantee of fuel supply.

However, foreign reactor suppliers have put most plans on hold due to the possible implications of India's Civil Liability for Nuclear Damage Act 2010. See Nuclear liability section below.

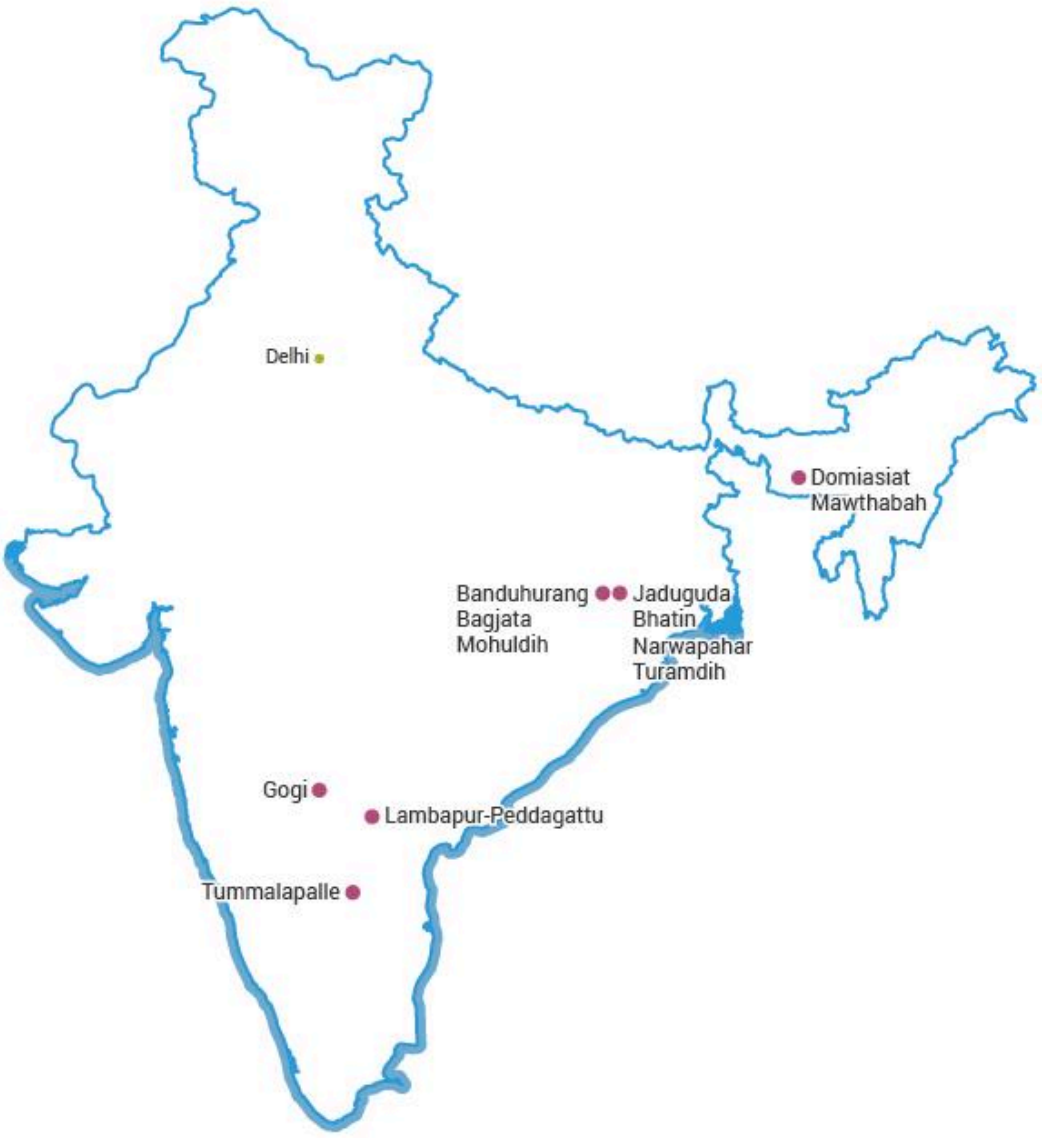
Uranium

Uranium resources and mining

India's uranium resources are modest, with 292,867 tonnes of uranium as identified resources in situ*, 282,401 tU of this as reasonably assured resources in situ and 10,466 tonnes as inferred resources in situ (to $260/kgU) at January 2021 in the OECD NEA 'Red Book'. In July 2017, 229,499 tU was clamied by the DAE. These are all in a high-cost category, and India expects to import an increasing proportion of its uranium fuel needs. In 2013 it was importing about 40% of uranium requirements. In July 2015 record annual domestic production of 1252 t U_3O_8 (1062 tU) was reported. However, 2015 production was only 385 tU.

* 56% carbonate deposits, 25% metamorphite, 7% sandstone, 6% proterozoic unconformity, 4% metasomatic, 2% granite-related, and <1% paleo-quartz-pebble-conglomerate.

Uranium mines in India



Exploration is carried out by the Atomic Minerals Directorate for Exploration and Research (AMD).

Mining and processing of uranium is carried out by Uranium Corporation of India Ltd (UCIL), also a subsidiary of the Department of Atomic Energy (DAE), in Jharkhand near Calcutta. Common mills are near Jaduguda (2500 t/day) and Turamdih (3000 t/day, expanding to 4500 t/day). Jaduguda ore is reported to grade 0.05-0.06%U. All Jharkhand mines are in the Singhbhum shear zone, and all are underground except Banduhurang. Another mill is at Tummalapalle in AP, expanding from 3000 to 4500 t/day.

In 2005 and 2006 plans were announced to invest almost $700 million to open further mines: in Jharkand at Banduhurang, Bagjata and Mohuldih; in Meghalaya at Domiasiat-Mawthabah (with a mill); and in Telangana at Lambapur-Peddagattu (with mill 50km away at Seripally), both in Nalgonda district.

Most production is via acid leach, producing magnesium diuranate or uranium peroxide via ion exchange. At Tummalapalle alkaline leach is used to produce sodium diuranate.

Uranium resources (as of July 2017)

State	Districts	Main deposits	Tonnes U
Andhra Pradesh	YSR	Tummalapalle	120,229
	Guntur	Koppunuru	2341
	Kurnool	Kappatralla	?
Telangana	Nalgonda	Lambapur, Pedagattu, Chitrial	15,731
Jharkhand	E.Singhbhum	Jaduguda, Bhatin, Narwapahar, Turamidh, Banduhurang, Mohuldih, Bagjata,	53,237
	Saraikela-Kharswan	Bangurdih	1367
Meghalaya	West Khasi Hills	KPM (Domiasat), Wahkyn, Wahkut	19,538
Rajasthan	Sikar, Udaipur	Rohil, Umra	7989
Karnataka	Yadgir, S.Kanara	Gogi	3970
Chhattisgarh	Rajanandgaon, Surguja	Bodal, Jajawal	3380
Uttar Pradesh	Sonbhadra	Naktu	666
Uttarakhand	Rudraprayag	Pokhri-Tunji	85
Himachal Pradesh	Una, Shimla, Mandi	Rajpura	665
Maharashtra	Gondia	Mogarra	301
Total			229,499

The **Jaduguda** mine in Jharkhand was closed in September 2014 due to expiry of its mining licence, but this was renewed a few weeks later by the state government, and in December the East Singhbhum government gave approval to resume mining, subject to clearance from the forestry department, which was still awaited at the end of 2015. The AMD quotes resources as 6816 tU (March 2014). Serious questions have been raised about health issues and environmental management.*

* To 2015, three separate health surveys were carried out by independent specialists. They concluded that alleged health effects were not caused by radiation. One medical team noted that the problems noted can be seen in any Indian village with similar socio-economic parameters. The radiation due to UCIL operations is negligible compared with natural background radiation.

In Jharkand, **Banduhurang** is India's first open cut mine and was commissioned in 2007. Bagjata is underground and was opened in December 2008, though there had been earlier small operations 1986-91. The **Mohuldih** underground mine was commissioned in April 2012. The new mill at Turamdih serving these mines was commissioned in 2008. It is 7 km from Mohuldih. Narwapahar and Bhatin are other underground mines in this area.

In Andhra Pradesh (AP) and Telangana there are three kinds of uranium mineralization in the Cuddapah Basin, including unconformity-related deposits in the north of it.

In the north of the Basin, in Telangana, the new northern inland state subdivided from Andhra Pradesh in 2013, the **Lambapur-Peddagattu** project in Nalgonda district 110 km southeast of Hyderabad has environmental clearance for one open cut and three small underground mines (based on some 6000 tU resources at about 0.1%U) but faces local opposition. The central government had approved Rs 637 crore for the project, with processing to be at Seripally, 54 km away in Nalgonda district. In 2014 UCIL was preparing to approach the state government and renew its federal approvals for the project. A further deposit near Lambapur-Peddagattu is Koppunuru, in Guntur district of AP, now under evaluation, and Chitrial.

In the south of the Basin, the **Tummalapalle** belt with low-grade strata-bound carbonate uranium mineralization is 160 km long, and appears increasingly prospective – the AMD reports 37,000 tU in 15 km of it and over 100,000 tU overall, extending down dip to 1000 metres. Some secondary mineralization is reported in the Srisailam sub-basin.

In August 2007 the government approved a new US$ 270 million underground mine and mill at Tummalapalle near Pulivendula in YSR (Cuddapah) district of Andhra Pradesh, 300 km south of Hyderabad. Its resources have been revised upwards by the AMD to 71,690 tU (March 2014) and its cost to Rs 19 billion ($430 million), and to the end of 2012 expenditure was Rs 11 billion ($202 million). First commercial production was in June 2012, using an innovative pressurized alkaline leaching process (this being the first

time alkaline leaching is used in India). Production is expected to reach 220 tU/yr as sodium diuranate, and in 2013 mill capacity was being doubled at a cost of Rs 8 billion ($147 million). An expansion of or from the Tummalapalle project is the Kanampalle U project, with 38,000 tU reserves. Further southern mineralization near Tummalapalle are Motuntulapalle, Muthanapalle, and Rachakuntapalle. The AMD has applied for a uranium prospecting licence for Kappatralla in the Kurnool district of AP, between the YSR district and Telangana.

In Karnataka, to the west of north Cuddapah Basin, UCIL is planning a small underground uranium mine in the Bhima basin at **Gogi** in Gulbarga area from 2014, after undertaking a feasibility study, and getting central government approval in mid-2011, state approval in November 2011 and explicit state support in June 2012. A portable mill is planned for Diggi or Saidpur nearby, using conventional alkaline leaching. Total cost is about $135 million. Resources are 4250 tU at 0.1% (seen as relatively high-grade) including 2600 tU reserves, sufficient for 15 years mine life, at 127 tU/yr, from fracture/fault-controlled uranium mineralization. UCIL plans also to utilize the uranium deposits in the Bhima belt from Sedam in Gulbarga to Muddebihal in Bijapur.

In Meghalaya, close to the Bangladesh border in the South West Khasi Hills, the **Kylleng-Pyndengsohiong-Mawthabah** – **KPM** – (known formerly as Domiasiat) mine project (near Nongbah-Jynrin) is on private land in a high rainfall area and has also faced longstanding local opposition partly related to land acquisition issues but also fanned by a campaign of fearmongering. For this reason, and despite earlier state government support in principle, UCIL was unable to get approval from the state government for the KPM open cut mine, though pre-project development had been authorized on 422 ha. Earlier there was federal environmental approval in December 2007 for a proposed uranium mine and processing plant here and for the Nongstin mine. There is sometimes violent opposition by NGOs to uranium mine development in the West Khasi Hills, including at KPM/Domiasiat and Wakhyn, which have estimated resources of 9500 tU and 8000 tU respectively. Tyrnai is a smaller deposit in the area. The status and geography of all these is not known, beyond the AMD being reported as saying that UCIL is "unable to mine them because of socio-economic problems." In August 2016 the Meghalaya state government cancelled UCIL's lease in the South West Khasi Hills district to deliver a “strong message” against uranium mining in the state. At the end of 2016 the AMD called for tenders for 15,000 metres of exploration core drilling on the Nongiri Plateau in the

South West Khasi Hills district. However in 2018 three AMD officials were assaulted whilst collecting uranium samples from small boreholes, prompting UCIL to close its project office in Meghalaya in August 2018 and terminate service agreements.

Fracture/fault-controlled uranium mineralization similar to that in Karnataka in the North Delhi Fold Belt is in the 130 km long Rohil belt in Sikar district in Rajasthan, with 6133 tU identified (March 2014).

The AMD reports further uranium resources in Chattisgarh state (3380 tU), Himachal Pradesh (665 tU), Maharashtra (300 tU), and Uttar Pradesh (750 tU).

In Jharkhand UCIL has a small project to recover uranium from copper tailings, near Hindustan Copper's Rakha and Surda mines.

India's uranium mines and mills – existing and planned

State, district	Mine	Mill	Operating from	tU per year
Jharkhand	Jaduguda	Jaduguda	1967 (mine) 1968 (mill)	200 total from mill
	Bhatin	Jaduguda	1967	
	Narwapahar	Jaduguda	1995	
	Bagjata	Jaduguda	2008	
Jharkhand, East Singhbum dist.	Turamdih	Turamdih	2003 (u/g mine) 2008 (mill)	190 total from mill
	Banduhurang	Turamdih	2007 (open pit)	
	Mohuldih	Turamdih	2012	
Andhra Pradesh, YSR district	Tummalapalle	Tummalapalle	2012 2015 (mill)	220 increasing to 330
Andhra Pradesh, YSR district	Tummalapalle	Kanampalle?	2017?	
Telangana, Nalgonda district	Lambapur-Peddagattu	Seripally/Mallapuram	2024? (open pit + 3 u/g)	130
Karnataka, Yadgir (Gulbarga) district	Gogi	Diggi/Saidapur	2020? (underground)	130

State, district	Mine	Mill	Operating from	tU per year
Meghalaya, West Khasi Hills district	Kylleng-Pyndeng-Sohiong-Mawthabah (KPM), (Domiasiat), Wakhyn	Mawthabah	suspended	340

However, India has reasonably assured resources of 319,000 tonnes of thorium – about 13% of the world total, and these are intended to fuel its nuclear power program longer-term (see below). The AMD claims almost 12 million tonnes of monazite which might contain 700,000 tonnes of thorium.

In September 2009 largely state-owned Oil & Natural Gas Corporation ONCC proposed to form a joint venture with UCIL to explore for uranium in Assam, and was later reported to be mining uranium in partnership with UCIL in the Cauvery area of Tamil Nadu.

Uranium imports

Following an IAEA safeguards agreement, an NSG resolution and finally US Congress approval of a bilateral trade agreement in October 2008, two months later Russia's Rosatom and Areva from France had contracted to supply uranium for power generation, while Kazakhstan, Brazil and South Africa were preparing to do so. The Russian agreement was to provide fuel for PHWRs as well as the two small Tarapur reactors.

In February 2009 the actual Russian contract was signed with TVEL to supply 2000 tonnes of natural uranium fuel pellets for PHWRs over ten years, costing $780 million, and 58 tonnes of low-enriched fuel pellets for the Tarapur reactors. The 300 tU Areva shipment arrived in June 2009. RAPS 2 became the first PHWR to be fuelled with imported uranium, followed by units 5&6 there.

In January 2009 NPCIL signed a memorandum of understanding with Kazatomprom for the supply of 2100 tonnes of uranium oxide concentrate (UOC) over six years and a feasibility study on building Indian PHWR reactors in Kazakhstan. NPCIL said it represented "a mutual commitment to begin thorough discussions on long-term strategic relationship." The actual agreement in April 2011 covered 2100 tonnes by 2014. In March 2013 both countries agreed to

extend the civil nuclear cooperation agreement past 2014. In 2015 the DAE renewed its contract for supply of 5000 tU from Kazatomprom over four years.

In September 2009 India signed uranium supply and nuclear cooperation agreements with Namibia and Mongolia. The latter was reaffirmed in May 2015, noting that Mongolian uranium "could help power India's low-carbon growth."

In March 2010 Russia offered India a stake in the Elkon uranium mining development in its Sakha Republic, and agreed on a joint venture with ARMZ Uranium Holding Co.

In August 2014 Navoi Mining and Metallurgical Combine (NMMC) in Uzbekistan signed a contract for supply of 2000 tonnes of U_3O_8 to India during the four years to 2018, its first export to India. A further contract was signed in January 2019, for long-term supply.

In September 2014 a bilateral safeguards agreement with Australia was signed, then came into force in November, enabling supply from there.

In April 2013 a bilateral safeguards agreement was signed between the DAE and the Canadian Nuclear Safety Commission (CNSC), and in April 2015 Cameco signed an agreement to supply 3200 tonnes of U_3O_8 (UOC) to India up to 2020. The first Cameco shipment arrived in December 2015.

In July 2015 the DAE reported to parliament that eight reactors (Kaiga 1-4, Madras 1&2 and Tarapur 3&4) were using indigenous sources of uranium and 14 reactors were using imported uranium. This situation was confirmed in July 2016 and July 2017.

In 2014 the DAE reported that India had imported 4458 tonnes of uranium since 2008 (2058 t from TVEL, 2100 t from Kazatomprom, and 300 t from Areva).

Uranium imports from 2014

Year	Source	Form	Tonnes
2014-15	TVEL Kazatomprom	UO_2 pellets UOC	297 283
2015-16	Cameco TVEL	UOC UO_2 pellets	251 346
2016-17	Kazatomprom Cameco TVEL	UOC UOC UO_2 pellets	1924 1234 187

Fuel cycle

India's main nuclear fuel cycle complex is at Hyderabad in Telangana, established in 1971. It plans to set up three more to serve the planned expansion of nuclear power and bring relevant activities under international safeguards. The first of the three will be at Kota in Rajasthan, supplying fuel for the 700 MWe PHWRs at Rawatbhata and Kakrapar by 2016. Capacity will be 500 t/yr plus 65 t of zirconium cladding. The second new complex will supply fuel to ten 700 MWe PHWRs planned in Haryana, Karnataka and Madhya Pradesh, but its site is not announced. The third will be at Chitradurga in the south of Karnataka state on a site with other science-based establishments, starting with a BARC enrichment plant, to supply fuel for light water reactors (see below).

DAE's Nuclear Fuel Complex (NFC) at Hyderabad has six facilities under safeguards, listed in the Annex to India's Additional Protocol with IAEA. This includes several facilities related to fuel fabrication, as part of the civil-military separation.

The NFC undertakes **refining and conversion** of uranium, which is received as magnesium diuranate (yellowcake) and refined to UO_2. The main 1250 t/yr plant fabricates PHWR fuel (which is unenriched). A small (25 t/yr) fabrication plant makes fuel for the Tarapur BWRs from imported enriched (2.66% U-235) uranium. Depleted uranium oxide fuel pellets (from reprocessed uranium) and thorium oxide pellets are also made for PHWR fuel bundles. Mixed carbide fuel for FBTR was first fabricated by Bhabha Atomic Research Centre (BARC) in 1979.

Heavy water is supplied by DAE's Heavy Water Board, and the seven plants have been working at capacity due to the current building program. Some $16 million worth of heavy water was exported to USA and France in 2013-14.

A very small centrifuge **enrichment** plant – insufficient even for the Tarapur reactors – is operated by DAE's Rare Materials Plant (RMP) at Ratnahalli near Mysore, primarily for military purposes including submarine fuel, but also supplying research reactors. It started up about 1992 as a unit of BARC, and is apparently being expanded

to some 25,000 SWU/yr. A conversion plant is also being built there at RMP.

Some centrifuge R&D is undertaken by BARC at Trombay.

DAE in 2011 announced that it would build an industrial-scale centrifuge complex, the Special Material Enrichment Facility (SMEF), in Chitradurga district, Karnataka, also as part of BARC and having both civil and naval purposes. Construction had not started in mid 2015. India's enrichment plants are not under international safeguards.

Fuel fabrication is by DAE's Nuclear Fuel Complex in Hyderabad. It services the Tarapur BWRs among others. This plant produces 1500 t/yr of PHWR fuel. DAE is setting up a second Nuclear Fuel Complex (NFC) – a PHWR fuel plant at Kota in Rajasthan, next to the Rawatbhata power plant – to serve the larger new reactors and those in northern India. It will have 500 t/yr capacity, from 2017, and government approval of Rs 2400 crore (24 billion rupees, $393 million) for this was in March 2014. Each 700 MWe reactor is said to need 125 t/yr of fuel. A third fuel fabrication plant is planned, with 1250 t/yr capacity, in Telangana, Rajasthan or Madhya Pradesh. The company is proposing joint ventures with US, French and Russian companies to produce fuel for those reactors.

Reprocessing: Used fuel from the civil PHWRs is reprocessed by Bhabha Atomic Research Centre (BARC) at Trombay, Tarapur and Kalpakkam to extract reactor-grade plutonium for use in the fast breeder reactors. The first 'plutonium plant' was commissioned in 1964 at Trombay, for weapons. Then the Power Reactor Fuel Reprocessing (PREFRE) facility at Tarapur was commissioned in 1979, and in 2010 a second PREFRE plant with 100 t/yr capacity effectively replaced it. A new Kalpakkam plant (KARP) of some 100 t/yr was commissioned in 1998 in connection with Indira Gandhi Centre for Atomic Research (IGCAR), though it was shut down over 2003-2009 due to an accident, then upgraded. It is being extended to reprocess FBTR carbide fuel. Apart from this all reprocessing uses the Purex process. A P3A project is being built to increase the capacity at Kalpakkam.

Partitioning of Purex product in a multi-step solvent extraction process is being undertaken in a demonstration facility at Tarapur. Civil plutonium initially has gone into the FBTR, the amount being estimated at 200-250 kg. More recently most has been for the PFBR, which is expected to require 400 kg/yr in full operation. India's civil plutonium stock at the end of 2014 is estimated at about 2.9 tonnes, mostly in connection with the PFBR.

Reprocessing capacity is understood to be about 100 t/yr at Tarapur and 100 t/yr at Kalpakkam, total 200 t/yr, but actually in operation about 115 t/yr producing 400 kg/yr plutonium, all related to the indigenous PHWR programme and not under international safeguards.

An away-from-reactor (AFR) fuel storage and another store at Tarapur are under safeguards from 2012 and 2014 and are listed in the AP Annex.

The Power Reactor Thoria Reprocessing Facility (PRTRF) was under construction at BARC in October 2013, and is designed to cope with high gamma levels from U-232. The recovered U-233 will be used in the AHWR Critical Facility.

India will reprocess the used PWR fuel from the Kudankulam and other imported reactors and will keep the plutonium. This will be under IAEA safeguards, in new plants.

In April 2010 it was announced that 18 months of negotiations with the USA had resulted in agreement to build two new reprocessing plants to be under IAEA safeguards, likely located near Kalpakkam and near Mumbai – possibly Trombay. In July 2010 an agreement was signed with the USA to allow reprocessing of US-origin fuel at one of these facilities. Since then the first **Integrated Nuclear Recycle Plant (INRP)** with facilities for both reprocessing of used light water reactor fuel of foreign origin, and waste management has been designed. Hindustan Construction Company (HCC) in October 2015 won the Rs 943 crore contract to build this at BARC at Tarapur. The plant will process used fuel from new nuclear power plants, including Gorakhpur 1&2 at Haryana, Rajasthan 7&8, Kakrapar 3&4 and future PHWRs.

In 2003 a facility was commissioned at Kalpakkam to reprocess mixed carbide fuel using an advanced Purex process. In 2010 the AEC said that used mixed carbide fuel from the Fast Breeder Test Reactor (FBTR) with a burn-up of 155 GWd/t was reprocessed in the Compact Reprocessing facility for Advanced fuels in Lead cells (CORAL). Thereafter, the fissile material was refabricated as fuel and loaded back into the reactor, thus 'closing' the fast reactor fuel cycle for the FBTR.

Fast Reactor Fuel Cycle Facility (FRFCF)

To close the main FBR fuel cycle the Fast Reactor Fuel Cycle Facility (FRFCF) has long been planned, with construction originally to

begin in 2008 and operation to coincide with the need to reprocess the first PFBR fuel. The PFBR will use oxide fuel. After that it is expected that metal fuel with higher breeding capability will be introduced and burnup is intended to increase from 100 to 200 GWd/t.

In July 2013 the government approved construction of the Rs 9600 crore (96 billion rupees, $1.61 billion) FRFCF at Kalpakkam. Work was expected to start in 2013, initially under the auspices of the Indira Gandhi Centre for Atomic Research (IGCAR). In August 2017 HCC was awarded an Rs 7.64 billion/764 crore ($120 million) contract by IGCAR for building the FRFCF over 48 months. Earlier associated contracts for HCC covered infrastructure at Kalpakkam and a metal fuel plant – the Demonstration Facility for Metallic Fuels (DFMF) contracted in March 2016 for Rs 43.15 crore. The DFMF is to be followed by a commercial plant with 50 times the capacity. The FRFCF was commissioned in 2024. It will serve the PFBR nearby, and has capacity to cater for three such reactors.

Thorium fuel cycle development in India

The long-term goal of India's nuclear program has been to develop an advanced heavy-water thorium cycle.The first stage of this employs the PHWRs fuelled by natural uranium, and light water reactors, which produce plutonium incidentally to their prime purpose of electricity generation.

Stage 2 uses fast neutron reactors burning the plutonium with the blanket around the core having uranium as well as thorium, so that further plutonium (ideally high-fissile Pu) is produced as well as U-233.

The AMD has identified almost 12 million tonnes of monazite resources (typically with 6-7% thorium) and 33.7 million tonnes of zircon.

Then in stage 3, Advanced Heavy Water Reactors (AHWRs) will burn thorium-plutonium fuels in such a manner that breeds U-233 which can eventually be used as a self-sustaining fissile driver for a fleet of breeding AHWRs. An alternative stage 3 is molten salt breeder reactors (MSBR), which are firming up as an option for eventual large-scale deployment. See R&D section under IGCAR.

In 2002 the regulatory authority issued approval to start construction of a 500 MWe prototype fast breeder reactor at Kalpakkam and this is now under construction by BHAVINI. It will

be fuelled with uranium-plutonium mixed oxide (MOX, the reactor-grade Pu being from its existing PHWRs). It will have a blanket with thorium and uranium to breed fissile U-233 and plutonium respectively. This will take India's ambitious thorium programme to stage 2, and set the scene for eventual full utilization of the country's abundant thorium to fuel reactors. Six more such 500 MWe fast reactors were announced for construction, four of them by 2020, but these have not yet proceeded. A fleet of fast reactors will breed the required plutonium which is the key to unlocking the energy potential of thorium in AHWRs. This will take another 15-20 years, and so it will still be some time before India is using thorium energy to any extent.

So far about one tonne of thorium oxide fuel has been irradiated experimentally in PHWR reactors* and some of this has been reprocessed, according to BARC.

* Notably Kakrapar 1&2, Rajasthan 2-4, Kaiga 1&2 have irradiated 232 fuel bundles to maximum burn-up of 14 GWd/t.

In October 2013 BARC said that premature deployment of thorium would lead to sub-optimal use of indigenous energy resources, and that it would be necessary to build up a significant amount of fissile material before launching the thorium cycle in a big way for the third stage (though the demonstration AHWR could be operating by 2022). Incorporation of thorium in the blankets of metal-fuelled fast breeder reactors would be after significant FBR capacity was operating. Hence thorium-based reactor deployment is expected to be “beyond 2070”. Surplus U-233 from FBR blankets could be used in HTRs including molten salt breeder reactors. See R&D section under IGCAR.

Design of the first 300 MWe AHWR (920 MWt, 284 MWe net) was completed early in 2014 at BARC. It is mainly a thorium-fuelled reactor but is versatile regarding fuel. Construction of the first one is due to start in the 12th plan period to 2017, for operation about 2022. At the end of 2016 large-scale engineering studies were validating innovative features of the design. No site or construction schedule had been announced for the demonstration unit. The AHWR can be configured to accept a range of fuel types including U-Pu MOX, Th-Pu MOX, and Th-U-233 MOX in full core, the U-233 coming from reprocessing in closed fuel cycle. A co-located fuel cycle facility is planned, with remote handling for the highly-radioactive fresh fuel.*

* In 2008 an AHWR critical facility was commissioned at BARC "to conduct a wide range of experiments, to help validate the reactor

physics of the AHWR through computer codes and in generating nuclear data about materials, such as thorium/uranium-233 based fuel, which have not been extensively used in the past." It has all the components of the AHWR's core including fuel and heavy water moderator, and can be operated in different modes with various kinds of fuel in different configurations.

The 300 MWe AHWR will have vertical pressure tubes in which the light water coolant under high pressure will boil at 285°C, circulation being by convection. Thermal efficiency is 30.9%. It is moderated by heavy water. There are 452 fuel assemblies, with burn-up of 38 GWd/t. A large heat sink or "gravity-driven water pool" with 7000 cubic metres of water is near the top of the reactor building and has a safety function. It has a slightly negative void coefficient of reactivity and several advanced passive safety features to enable meeting next-generation safety requirements such as 72-hour grace period for operator response, elimination of the need for exclusion zone beyond the plant boundary, 100-year design life, and high level of fault tolerance. The advanced safety characteristics have been verified in a series of experiments carried out in full-scale test facilities. It is claimed that per unit of energy produced, the amount of long-lived minor actinides generated is nearly half of that produced in current generation light water reactors. A high level of radioactivity in the fissile and fertile materials recovered from the used fuel of the AHWR, and their isotopic composition, preclude the use of these materials for nuclear weapons*.

* 9.5% of the plutonium is Pu-238.

In 2009 the AEC also announced an export version of the AHWR, the **AHWR300-LEU.** This will use low-enriched uranium plus thorium (Th-LEU MOX) as a fuel, dispensing with the plutonium input. About 39% of the power will come from thorium (via in situ conversion to U-233, cf two-thirds in AHWR), and burn-up will be 61 GWd/t. Uranium enrichment level will be 19.75%, giving 4.21% average fissile content of the U-Th fuel. The design is based on once-through fuel cycle during its lifetime. While closed fuel cycle is possible, this is not required or envisaged, and the used fuel, with about 8% fissile isotopes can be used in light water reactors. Plutonium production will be less than in light water reactors, the fissile proportion will be less and the Pu-238 portion three times as high. With also a significant level of gamma-emitting U-232 in the used fuel, there is inherent proliferation resistance. The design is intended for overseas sales, and the AEC says that "the reactor is manageable with modest industrial infrastructure within the reach of developing countries".

A third variety is the AHWR-Pu, which will have Pu-Th MOX and Th-U-233 MOX fuel.

An NPCIL presentation early in 2012 had LEU AHWRs being fueled with LEU-thorium, while U-233 and thorium from fast reactors, along with used fuel from those AHWRs, fueled accelerator-driven subcritical molten salt reactors. Thorium was evidently the main fuel for both these types. Also AHWR-LEU produces half as much minor actinides as LWR.

The conceptual design of an Indian Molten Salt Breeder Reactor (IMSBR) has been commenced. No details are announced.

Radioactive waste management

In October 2013 BARC stressed the role of accelerator-driven subcritical molten salt reactor systems (ADS) burning minor actinides arising from partitioning of PHWR and LWR Purex output. These working in tandem would address waste issues more effectively and safely than using critical fast reactors to burn minor actinides. Pyroprocessing would treat these wastes.

Radioactive wastes from the nuclear reactors and reprocessing plants are treated and stored at each site. Waste immobilization plants (WIP) are in operation at Tarapur and Trombay and another vitrification plant was commissioned by BARC in 2013 at Kalpakkam for wastes from reprocessing Madras (MAPS) used fuel. The WIPs use borosilicate glass, as in Europe.

Research on final disposal of high-level and long-lived wastes in a geological repository is in progress at BARC.

Heavy engineering in India

India's largest engineering group, Larsen & Toubro (L&T) announced in July 2008 that it was preparing to venture into international markets for the supply of heavy engineering components for nuclear reactors. It formed a 20 billion rupee ($463 million) venture with NPCIL to build a new plant for domestic and export nuclear forgings at its Hazira, Surat coastal site in Gujarat state. This would produce 600-tonne ingots in its steel melt shop and have a very large forging press to supply finished forgings for nuclear reactors, pressurizers and steam generators, and also heavy forgings for critical equipment in the hydrocarbon sector and for thermal power plants. In 2015 Westinghouse said that it was equipped to produce

reactor pressure vessels and other major components for AP1000 reactors.

In the context of India's trade isolation over three decades L&T has produced heavy components for 17 of India's pressurized heavy water reactors (PHWRs) and has also secured contracts for 80% of the components for the fast breeder reactor at Kalpakkam. It is qualified by the American Society of Mechanical Engineers to fabricate nuclear-grade pressure vessels and core support structures, achieving this internationally recognized quality standard in 2007, and further ASME accreditation in 2010. It is one of about ten major nuclear-qualified heavy engineering enterprises worldwide.

Early in 2009, L&T signed four agreements with foreign nuclear power reactor vendors. The first, with Westinghouse, set up L&T to produce component modules for Westinghouse's AP1000 reactor. The second agreement was with Atomic Energy of Canada Ltd (AECL) "to develop a competitive cost/scope model for the ACR-1000" (though this would have lapsed). In April it signed an agreement with Atomstroyexport primarily focused on components for the next four VVER reactors at Kudankulam, but extending beyond that to other Russian VVER plants in India and internationally. Then in May 2009 it signed an agreement with GE Hitachi (GEH) to produce major components for ABWRs from its new Hazira JV plant. The two companies hope to utilize indigenous Indian capabilities for the complete construction of nuclear power plants including the supply of reactor equipment and systems, valves, electrical and instrumentation products for ABWR plants to be built in India. L&T "will collaborate with GEH to engineer, manufacture, construct and provide certain construction management services" for an envisaged ABWR project, possibly at Chutka in Madhya Pradesh. Early in 2010 L&T signed an agreement with Rolls-Royce to produce technology and components for light water reactors in India and internationally.

Following the 2008 removal of trade restrictions, Indian companies led by Reliance Power (RPower), NPCIL, and Bharat Heavy Electricals Ltd (BHEL) said that they planned to invest over $50 billion in the next five years to expand their manufacturing base in the nuclear energy sector. BHEL planned to spend $7.5 billion in two years building plants to supply components for reactors of 1600 MWe. Mumbai-based Walchandnagar Industries Limited aims to build a factory in Gujarat in joint venture with Atomenergomash OJSC in line with a 2010 agreement to build nuclear power equipment for both Indian and export markets. If this did not proceed, it was open to a JV with Westinghouse or EDF.

BHEL also planned to set up a tripartite joint venture with NPCIL and Alstom to supply turbines for nuclear plants of 700 MWe, 1000 MWe and 1600 MWe. In June 2010 Alstom confirmed that the equal joint venture with NPCIL and BHEL would be capitalized to €25 million, to provide turbines initially for eight 700 MWe PHWR units, then for imported large units. Another joint venture is with NPCIL and a foreign partner to make steam generators for 1000-1600 MWe plants.

Two contracts awarded by NPCIL to a consortium of BHEL and Alstom cover the supply and installation of turbogenerator packages for Kakrapar 3&4, the first indigenously designed 700 MWe pressurized heavy water reactors. The contracts are worth over INR 16,000 million ($360 million), with BHEL's share representing around INR 8000 million ($200 million). The first contract covers the supply of the actual turbine generator packages, while the second covers associated services. BHEL and Alstom would jointly manufacture and supply the steam turbines, while BHEL will manufacture and supply the generator, moisture separator reheater and condenser, as well as undertaking the complete erection and commissioning of the turbine generator package. In August 2012 NPCIL awarded an INR 19,060 million ($340 million) contract to BHEL-Alstom for turbine generators for Rajasthan 7&8. Under the contract, BHEL and Alstom would together manufacture and supply the steam turbines, while the manufacture and supply of the complete generator, moisture separator reheater and condenser – including the erection and commissioning of the turbine generator package – will be undertaken by BHEL.

BHEL is also supplying steam generators for one Kakrapar unit and Rajasthan 7&8. It will also supply and install the instrumentation and control systems for the turbine island secondary circuit for Rajasthan 7&8. BHEL is also supplying, constructing and commissioning the complete conventional island for the 500 MWe prototype fast breeder reactor being built at Kalpakkam.

HCC (Hindustan Construction Co.) has built more than half of India's nuclear power capacity, notably all six units of the Rajasthan Atomic Power Project and also Kudankulam. It has an INR 8880 million ($160 million) contract for the main civil works for Rajasthan 7&8. It specializes in prestressed containment structures for reactor buildings. In September 2009 it formed a joint venture with UK-based engineering and project management firm AMEC to undertake consulting services and nuclear power plant construction. HCC has had major orders from NPCIL, and in 2017 was awarded an INR 7.64 billion ($120 million) contract from

IGCAR for the FRFCF reprocessing plant for fast reactors at Kalpakkam.

Areva signed an agreement with Bharat Forge Ltd in January 2009 to set up a joint venture in casting and forging nuclear components for both export and the domestic market, by 2012. BHEL expects to join this, and in June 2010 the UK's Sheffield Forgemasters became a technical partner with BHEL in a £30 million deal. The partners have shortlisted Dahej in Gujarat, and Krishnapatnam and Visakhapatnam in Andhra Pradesh as possible sites.

In August 2010 GE Hitachi Nuclear Energy (GEH) signed a preliminary agreement with India's Tata Consulting Engineers, Ltd. to explore potential project design and workforce development opportunities in support of GEH's future nuclear projects in India – notably the proposals for six ESBWR units – and around the world.

In April 2012 Atomenergomash was negotiating with potential Indian partners on localization of some production and design of equipment for nuclear power plants being built with Russian technology both in India and other Asian countries such as Bangladesh, Sri Lanka and Vietnam. In 2010 a memorandum of understanding with Walchandnagar Industries Ltd (India) had been signed by Atomenergomash. A strategic visions agreement signed in December 2014 for strengthening cooperation in atomic energy with Russia mentioned that the two countries would explore "opportunities for sourcing materials, equipment and services from Indian industry for the construction of the Russian designed nuclear power plants in third countries." In January 2016 the prime minister announced cooperation with Russia to increase Indian manufacturing content in future VVER reactors in India. In October 2016 the government and Rosatom reaffirmed the "active engagement of Indian manufacturing industry for local manufacturing in India of equipment and components for upcoming and future Russian-designed nuclear power projects."

In November 2018 Holtec International subsidiary Holtec Asia signed an MoU with the government of Maharashtra to establish a heavy manufacturing facility to support India's planned nuclear generation expansion. The $680 million Holtec Heavy Manufacturing Division plant would primarily fabricate complex and safety-related equipment for nuclear power plants but will also be equipped to meet the heavy welding needs of petroleum, chemical, aerospace and other industries. It is also intended to be used in the deployment of Holtec's SMR-160 reactor internationally.

See also India section of Heavy Manufacturing of Power Plants.

Research & Development

An early AEC decision was to set up the Bhabha Atomic Research Centre **(BARC)** at Trombay near Mumbai. A series of 'research' reactors and critical facilities was built here: APSARA (pool-type, 1 MW, operating 1956-2010) was the first research reactor in Asia, CIRUS (40 MWt, 1960) built under the Colombo Plan, and Dhruva (100 MWt, 1985) followed it along with fuel cycle facilities. CIRUS used natural uranium fuel, was moderated by heavy water and cooled by light water. It was extensively refurbished and then recommissioned in 2002, and ran to 2010. Dhruva was fully designed and built indigenously, and uses metallic uranium fuel with heavy water as moderator and coolant. Dhruva is extensively used in neutron beam research studies involving material science and nuclear fission processes. As well as research uses, the CIRUS and Dhruva units are assumed to be largely for military purposes, as is the Trombay plutonium plant commissioned in 1965. In line with international agreement, the government shut down CIRUS at the end of 2010.

Reprocessing of used fuel was first undertaken at Trombay in 1964. When opening the new reprocessing plant at Tarapur in 2011, the prime minister reminded listeners: "The recycling and optimal utilization of uranium is essential to meet our current and future energy security needs." The Actinide Separation Demonstration Facility is operated by BARC at Tarapur, to prepare the way for fissioning minor actinides in the fast reactors.

BARC is also responsible for the transition to thorium-based systems and in particular is developing the 300 MWe AHWR as a technology demonstration project. This will be a vertical pressure tube design with heavy water moderator, boiling light water cooling with passive safety design and thorium-plutonium based fuel (described more fully above). A large critical facility to validate the reactor physics of the AHWR core has been commissioned at BARC, and BARC's research laboratory at Tarapur tests various AHWR systems. An engineering-scale Power Reactor Thorium Reprocessing Facility (PRTRF) has been constructed at Trombay to reprocess thoria fuel bundles irradiated in PHWRs.

BARC is responsible for India's uranium enrichment projects, the pilot Rare Materials Plant (RMP) at Ratnahalli near Mysore, and the planned Special Material Enrichment Facility (SMEF) at Karnataka.

Zerlina was a 100-watt experimental reactor running 1961-83 using natural uranium fuel and heavy water moderator to test concepts for PHWRs.

On the occasion of signing a Canadian uranium supply agreement with NPCIL in April 2015 (based on the 2013 nuclear cooperation agreement with Canada), there was a joint prime ministerial agreement to encourage a collaborative programme to "leverage their industries' respective strengths" in pressurized heavy water reactor (PHWR) technology. It also encourages Canadian and Indian atomic energy establishments and research institutions to establish mechanisms for long-term collaboration in nuclear energy R&D, which would be centred at BARC. It includes an agreement to exchange nuclear safety and regulatory experiences and developments.

A series of three Purnima research reactors have explored the thorium cycle, the first (1971) running on plutonium fuel fabricated at BARC, the second and third (1984&1990) on U-233 fuel made from thorium – U-233 having been first separated in 1970. All three are now decommissioned. Thoria fuel rods irradiated in CIRUS have been reprocessed at the Uranium-Thorium Separation Facility (UTSF) at BARC with the recovered U-233 being fabricated as fuel for the Kamini reactor at IGCAR.

BARC has also designed an indigenous 900 MWe PWR, the Indian Pressurized Water Reactor (IPWR), which is to be deployed in collaboration with NPCIL. This follows its work building an 83 MW PWR at Kalpakkam for the *INS Arihant* submarine, which achieved criticality in mid-2013, using 40% enriched fuel. A 20 MW prototype submarine reactor operated at Kalpakkam from 2003 for several years. A second nuclear submarine, the *INS Arighat* (originally named *INS Aridhaman*), was launched in November 2017 and was commissioned in 2024.

In 1998 a 500 keV accelerator was commissioned at BARC for research on accelerator-driven subcritical systems (ADS) as an option for stage three of the thorium cycle.

The Raja Ramanna Centre for Advanced Technology (RRCAT) has Indus 1&2 synchrotrons operating – Indus 1 at 450 MeV and 100 mA, Indus 2 at 2.5 GeV and 150 mA though it has reached 200 mA.

There are plans for a new 20 MWt multi-purpose research reactor (MPRR) for radioisotope production, testing nuclear fuel and reactor materials, and basic research. It would use fuel enriched to 19.9% U-235 and is to be capable of conversion to an accelerator-driven system later.

Design studies are proceeding for a 200 MWe PHWR accelerator-driven system (ADS) fuelled by natural uranium and thorium. Uranium fuel bundles would be changed after about 7 GWd/t burn-up, but thorium bundles would stay longer, with the U-233 formed adding reactivity. This would be compensated for by progressively replacing some uranium with thorium, so that ultimately there is a fully-thorium core with in situ breeding and burning of thorium. This is expected to mean that the reactor needs only 140 tU through its operating lifetime and achieves a high burn-up of thorium – about 100 GWd/t. The disadvantage is that a 30 MW accelerator is required to run it.

The Indira Gandhi Centre for Atomic Research (IGCAR) at Kalpakkam was set up in 1971. Two civil research reactors here are preparing for stage two of the thorium cycle. BHAVINI is located here and draws upon the centre's expertise and that of NPCIL in establishing the fast reactor program, including the Fast Reactor Fuel Cycle Facility.

The 40 MWt fast breeder test reactor (FBTR) based on the French Rapsodie FBR design has been operating since 1985. It has achieved 165 GWday/tonne burn-up with its carbide fuel (70% PuC + 30% UC) without any fuel failure. In 2005 the FBTR fuel cycle was closed, with the reprocessing of 100 GWd/t fuel – claimed as a world first. This has been made into new mixed carbide fuel for FBTR. Prototype FBR fuel which is under irradiation testing in FBTR has reached a burn-up of 90 GWd/tonne. As part of developing higher-burn-up fuel for PHWRs, mixed oxide (MOX) fuel is being used experimentally in FBTR, which has been operating with a hybrid core of mixed carbide and mixed oxide fuel (the high-Pu MOX forming 20% of the core).

In 2011 FBTR was given a 20-year lifetime extension, to 2030, and IGCAR said that its major task over this period would be large-scale irradiation of the advanced metallic fuels and core structural materials required for the next generation fast reactors with high breeding ratios (the PFBR uses MOX fuel, but later versions will use metal.).

A 300 MWt, 150 MWe fast breeder reactor as a test bed for using metallic fuel is envisaged once several MOX-fuelled fast reactors

are in operation. This successor to FBTR will use U-Pu alloy or U-Pu-Zr, with electrometallurgical reprocessing. Its design is to be completed by 2017.

Also at IGCAR, the tiny Kamini (Kalpakkam mini) reactor is exploring the use of thorium as nuclear fuel, by breeding fissile U-233. It is the only reactor in the world running on U-233 fuel, according to the DAE.

The Compact High-Temperature Reactor (CHTR) of 100 kWt is being designed to have a long (15-year) core life and employ liquid metal (Pb-Bi eutectic) coolant. It uses TRISO fuel in tubes and blocks and is designed to operate at 1000°C for long periods giving high burn-up. It has a ceramic core with BeO and graphite moderator. It has several passive systems for heat removal. It is envisaged as a nuclear battery in remote areas with no grid.

The Innovative HTR (IHTR) of 600 MWt is envisaged for hydrogen production. It also uses TRISO fuel, with 7.3% U-233 at 1000°C, but in some 150,000 pebbles, hence online refuelling. It has active and passive systems for control and cooling. The molten salt coolant circulates by convection during normal operation. It is expected to produce 18 MWe and 80,000 m^3/h of hydrogen.

Also in the HTR area is conceptual design of the Indian Molten Salt Breeder Reactor (IMSBR) of 850 MWe which has potential to be used in stage 3 of the thorium programme. It would have a breeding ratio of 1.06 to 1.14 while operating in thermal or epithermal spectrum. The fissile inventory in a 850 MWe reactor would be almost 1 tonne, compared with 6 tonnes for metal-fuelled FBR, assuming online reprocessing. It has emphasis on passive systems for reactor heat removal under all scenarios and conditions. This is based on a French molten salt fast reactor concept.

The Board of Radiation & Isotope Technology (BRIT) was separated from BARC in 1989 and is responsible for radioisotope production. The research reactors APSARA, CIRUS and Dhruva are used, along with RAPS for cobalt-60. A regular supply of isotopes for various uses commenced in early 1960s after CIRUS became operational. At present the reactors supply some 1250 user institutions with preparations of Mo-99 , I-131 , I-125, P -32 , S-35, Cr-51 , Co-60, Au-198, Br-82, Ir-192 and others.

BARC has used nuclear techniques to develop 37 genetically-modified crop varieties for commercial cultivation. A total of 15 sterilising facilities, particularly for preserving food, are now operational with more under construction. Radiation technology has

also helped India increase its exports of food items, including to the most developed markets in the world.

India's hybrid Nuclear Desalination Demonstration Plant (NDDP) at Kalpakkam, comprises a reverse osmosis (RO) unit of 1.8 million litres per day commissioned in 2002 and a multi-stage flash (MSF) desalination unit of 4.5 million litres per day, as well as a barge-mounted RO unit commissioned recently, to help address the shortage of water in water-stressed coastal areas. It uses about 4 MWe from the Madras nuclear power station.

The Raja Ramanna Centre for Advanced Technology is a DAE unit engaged in R&D in non-nuclear frontline research areas of lasers, particle accelerators and related technologies. It runs the Indus 1&2 beamlines. The Variable Energy Cyclotron Centre is another DAE unit, specializing in accelerator science and technology, associated with BARC.

A new **Global Centre for Nuclear Energy Partnership** (GCNEP) was inaugurated in January 2014, pursuant to a September 2010 government approval. It will be the DAE's sixth R&D facility. It is being built near Bahadurgarh in Haryana state, 45km from Delhi airport, and designed to strengthen India's collaboration internationally. It will house five schools to conduct research into advanced nuclear energy systems, nuclear security, radiological safety, nuclear material characterization, as well as applications for radioisotopes and radiation technologies. Russia is to help set up four of the GCNEP schools. In March 2017 the IAEA agreed to provide staff for the centre and use it for training professionals throughout the region.

The DAE's **Atomic Minerals Directorate** for Exploration and Research (AMD) is focused on mineral exploration for uranium and thorium. It was set up in 1949, and is based in Hyderabad, with over 2700 staff. See also Mining section above.

Regulation, safety and non-proliferation

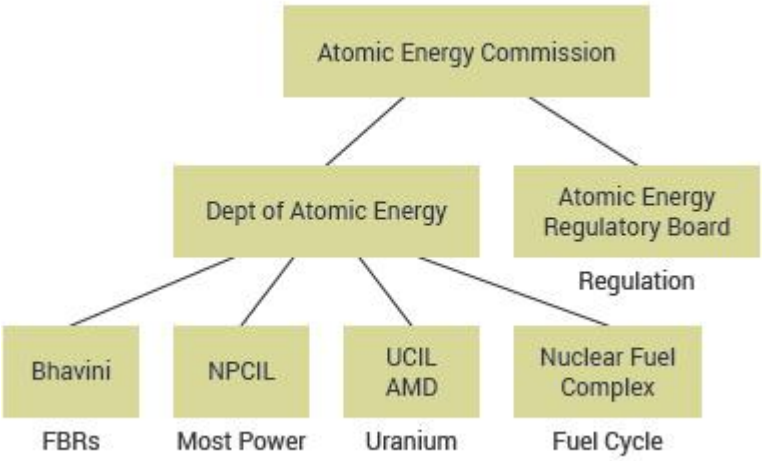
The Atomic Energy Commission (AEC) was established in 1948 under the Atomic Energy Act as a policy body. Then in 1954 the Department of Atomic Energy (DAE) was set up to encompass research, technology development and commercial reactor operation. The current Atomic Energy Act is 1962, and it permits only government-owned enterprises to be involved in nuclear power.

The DAE includes NPCIL, Uranium Corporation of India Ltd (UCIL, mining and processing), Atomic Minerals Directorate for Exploration and Research (AMD, exploration), Electronics Corporation of India Ltd (reactor control and instrumentation) and BHAVINI* (for setting up fast reactors). The DAE also controls the Heavy Water Board for production of heavy water and the Nuclear Fuel Complex for fuel and component manufacture.

* Bhartiya Nabhikiya Vidyut Nigam Ltd

The **Atomic Energy Regulatory Board (AERB)** was formed in 1983 and comes under the AEC but is independent of DAE. It is responsible for the regulation and licensing of all nuclear facilities, and their safety and carries authority conferred by the Atomic Energy Act for radiation safety and by the Factories Act for industrial safety in nuclear plants. However, it is not an independent statutory authority, and its 1995 report on a safety assessment of DAE's plants and facilities was reportedly shelved by the AEC. In April 2011 the government announced that it would legislate to set up a new independent and autonomous Nuclear Regulatory Authority of India that will subsume the AERB, and that previous safety assessments of Indian plants would be made public.

India Structure



Source: World Nuclear Association

In August 2012 a parliamentary report from the Comptroller and Auditor General (CAG) on the AERB pointed out serious organizational flaws and numerous failings relative to international norms. The most fundamental issue highlighted by the report was the unsatisfactory legal status and authority of the AERB. Despite India's international commitments, awareness of best practice and internal expert recommendations, the report said, "the legal status of AERB continued to be that of an authority subordinate to the central government, with powers delegated to it by the latter." The

CAG report emphasized the need to make the regulator independent of industry and government and insulated from commercial or political interference. The AERB had failed to prepare an overall nuclear radiation safety planning policy as required in 1983, and had failed to set up radiation safety directorates in 35 administrative areas to ensure the safe use of radiation in medical and industrial facilities, as required by a 2001 Supreme Court order. It had undertaken only 15% of the recommended level of inspections at industrial radiography and radiotherapy units, relative to IAEA norms, and had not achieved cost recovery from licensees. There was no detailed inventory of radioactive sources to help ensure safe disposal, and no "proper mechanism" to check the safe disposal of radioactive wastes.

This was largely anticipated and in September 2011 a bill to set up new stronger and more independent national nuclear regulatory authorities to oversee radiation and nuclear safety was introduced to India's lower house, the Lok Sabha. The Nuclear Safety Regulatory Authority Bill was drawn up in response to events at Fukushima and aimed to establish several new regulatory bodies. A new senior **Council of Nuclear Safety (CNS)** chaired by the prime minister would oversee and review policies on radiation safety, nuclear safety and other connected matters. It will include various government ministers, with the cabinet secretary and head of the Indian Atomic Energy Commission, plus government-nominated "eminent experts".

The second major body to be established was the **Nuclear Safety Regulatory Authority (NSRA)** to be responsible for ensuring radiation safety and nuclear safety in all civilian sector activities. The NSRA would take over the functions of the existing AERB. The bill lapsed and the government expected to reintroduce it in a 2013 session of parliament, but it was still pending at the end of 2015. The AEC invited the International Atomic Energy Agency's (IAEA) Integrated Regulatory Review Service (IRRS) to examine the new regulatory system, which will get statutory status after the passage of the Nuclear Safety Regulatory Authority (NSRA) Bill by Parliament. The IRRS reported favourably in March 2015, but said that “the Government should embed the AERB's regulatory independence in law, separated from other entities having responsibilities or interests that could unduly influence its decision making.” The IRRS review was led by a senior Canadian regulator, and by then the AERB had finalized an arrangement for regulatory cooperation in the field of nuclear and radiation safety regulation with the Canadian Nuclear Safety Commission (CNSC).

In 2012 an IAEA Operational Safety Review Team (OSART) reviewed the Rajasthan nuclear power plant, notably units 3&4, and reported favourably.

In April 2012 India's AERB joined the OECD Nuclear Energy Agency's Multinational Design Evaluation Program (MDEP) as its eleventh member, and first new member since the program's inception. The NEA said that it would be actively involved in the Codes and Standards Working Group, the Digital Instrumentation and Control Working Group, the Vendor Inspection Co-operation Working Group and, "eventually, one of the specific reactor design working groups." MDEP was launched in 2006 by the US NRC and France's ASN with the aim of coordinating national nuclear regulatory reviews of new power reactor designs.

In February 2015 the government signed a nuclear cooperation agreement with Sri Lanka. It is concerned with capacity building and training in peaceful application of nuclear energy, especially the use of radioisotopes, nuclear safety, radioactive waste management, radiation safety and nuclear security.

NPCIL is an active participant in the programs of the World Association of Nuclear Operators (WANO).

Nuclear liability

India's 1962 Atomic Energy Act says nothing about liability or compensation in the event of an accident. Also, India was not a party to the relevant international nuclear liability conventions (the IAEA's 1997 Amended Vienna Convention and 1997 Convention on Supplementary Compensation for Nuclear Damage – CSC). Since all civil nuclear facilities are owned and must be majority-owned by the Central Government (NPCIL and BHAVINI, both public sector enterprises), the liability issues arising from these installations are its responsibility. Following internal discussion on which might be the most appropriate international liability convention, on 10 September 2008 the government assured the USA that India "shall take all steps necessary to adhere to the Convention on Supplementary Compensation (CSC)". This requires domestic legislation which is consistent with it, but under existing Indian legislation, foreign suppliers faced potentially unlimited liability, which prevented them from taking insurance cover, though contracts for Kudankulam 1&2 excluded this supplier liability.

The Civil Liability for Nuclear Damage Act related to third party liability was passed by both houses of parliament in August 2010.

This is framed and was debated in the context of strong national awareness of the Bhopal disaster in 1984, probably the world's worst industrial accident. (A Union Carbide (51% US-owned) chemical plant in the central Madhya Pradesh state released a deadly mix of methyl isocyanate and other gases due to operator error and poor plant design, killing some 15,000 people and badly affecting some 100,000 others. The company paid out some US$ 1 billion in compensation – widely considered inadequate.)

The 2010 Act places responsibility for any nuclear accident with the operator, as is standard internationally, and limits total liability to 300 million SDR (about $450 million) "or such higher amount that the Central Government may specify by notification". Operator liability is capped at Rs 1500 crore (15 billion rupees, about $285 million) or such higher amount that the Central Government may notify, beyond which the Central Government is liable.

However, after compensation has been paid by the operator (or its insurers), clause 17(b) of the bill allows the operator to have legal recourse to the supplier for up to 80 years after the plant starts up if in the opinion of an Indian court the "nuclear incident has resulted as a consequence of an act of supplier or his employee, which includes supply of equipment or material with patent or latent defects or sub-standard services." This clause giving recourse to the supplier for an operational plant is contrary to international conventions and undermines the channeling principle fundamental to nuclear liability internationally. Also, no limit is set on suppliers' liability. The supplier community interpreted this provision as ambiguous and one that rendered it vulnerable to open-ended liability claims. A new explanation seeks to address it by relating Section 17(b) to 'actions and matters such as product liability stipulations/conditions or service contracts' between the operator and the supplier and therefore to be dealt with in the context of such contractual terms. The attempt is to remove the open-ended nature of possible liability claims by limiting these to the terms and conditions of the contract.

A second sticking point is Section 46 which states that the provisions of the Act 'were in addition to, and not in derogation of, any other law for the time being in force', leading to concerns among the suppliers that they could be subjected to multiple and concurrent liability claims. This is sought to be addressed by explaining that all civil claims can only be brought under the Act since that was the intention behind this special legislation and further, that these claims would come under the jurisdiction of the specially constituted Claims Commission, thereby excluding any

jurisdiction of foreign courts. And arguably clause 46 of the Act reinforces operator liability.

In November 2011 the DAE published a notification that claims by plant operators against component suppliers "shall in no case exceed the actual amount of compensation" paid by utilities. The new Civil Liability for Nuclear Damage Rules give plant operators the right of recourse against equipment suppliers related to "the extent of the operator's liability" or "the value of the contract itself, whichever is less." They also limit it to the duration of the initial plant licence "or the product liability period, whichever is longer." This is generally seen as confusing, and is not satisfactory to major suppliers, including Indian ones such as Larsen & Toubro.

It was reported that negotiations with Russia for additional nuclear reactors at Kudankulam were proceeding with an escalation of price because of this vendor liability sub-clause, in this case involving Atomstroyexport. The original Kudankulam agreement said that supplier liability ended with delivery of the plant. US diplomatic sources are similarly opposed to supplier liability after delivery, and GE-H, Westinghouse and Areva (now Framatome) sought changes to the law allowing vendor liability. Westinghouse said it would await India's ratification of the CSC before offering to supply equipment to India.

The bill does not make any mention of India ratifying the CSC or any international treaty or framework governing nuclear liability under which the supplier cannot be sued in their home country. The CSC was not yet in force internationally, but Indian ratification would bring it closer to being so, and was part of the September 2008 agreement with the USA. In October 2010 India signed the CSC. In 2011 the US Secretary for State said she expected India to ratify the CSC by year end, "and we would encourage engagement with the International Atomic Energy Agency to ensure that the liability regime that India adopts by law fully conforms with the international requirements under the convention." Eventually, in February 2016, India deposited its instrument of ratification of the CSC with the IAEA. However, it is not clear how it relates to the nuclear liability law, though the Ministry of External Affairs said in a statement that ratification of the CSC marked a "conclusive step in the addressing of issues related to civil nuclear liability in India." The US Energy Secretary welcomed the ratification. The CSC entered into force for India in May 2016.

Earlier, in April 2025 the Indian government created committees with members from the DAE, the AERB, the Ministry of Justice and the National Institution for Transforming India (NITI) to propose

amendments to the Atomic Energy Act and the Civil Liability for Nuclear Damage Act that would clarify matters of liability.

In May 2025 the country's Minister for Power set out the steps needed to scale the country's nuclear energy capacity, which included amending the 1962 Atomic Energy Act and 2010 Civil Liability for Nuclear Damage Act to enable broader participation by private and state sectors. In December 2025 both the lower and upper houses of Parliament passed the Sustainable Harnessing and Advancement of Nuclear Energy for Transforming India (SHANTI) Bill. The legislation repeals the 1962 Atomic Energy Act and the 2010 Civil Liability for Nuclear Damage Act and replaces them with a single, comprehensive law aligned with India's present and future energy requirements, according to the Department of Atomic Energy.

Funding the liabilities

In October 2010 it was reported that NPCIL proposed to set up a fund of Rs 1500 crore ($250 million) for nuclear liability "with the Centre addressing anything over this level".

In October 2013 it was reported that DAE had set up two committees to find a middle path, with a more "scientific and rational" approach to the issue. "The committees will assess the probabilistic safety analysis to identify a model that will assess probabilities of particular equipment or a set of system to fail in a manner that can lead to an accident. Based on the study there would be a rational basis for working out an actuarial approval to decide on the quantum of liability," according to DAE. The main committee comprises representatives from BARC, IGCAR and NPCIL (2). At the end of October 2013 the Planning Commission said that under the 2010 law the domestic plant operator could limit the amount as well as duration of the liability that accrues to foreign suppliers, so that the liability is limited and therefore insurable. However this interpretation is viewed with some scepticism.

In March 2014 the government reached some sort of agreement with Russia to provide liability insurance through the government-owned General Insurance Corporation of India (GIC), though the actual arrangements for a nuclear liability insurance product had yet to be worked out. GIC apparently discussed reinsurance with international companies, but without any agreement, due partly to the unlimited provisions of the 2010 Act, so was unable to proceed.

In April 2014 DAE approached the Ministry of Finance to urge the setting up of an **Indian nuclear insurance pool** as a high priority,

since insurance risks for third party liability alone amount to Rs 1500 crore. NIAEP-ASE, contracted to supply Kudankulam units 3&4, has insisted on the government providing reinsurance. In September 2014 the DAE and Ministry of Finance asked the GIC again to contrive a model for circumventing the right of recourse under the Civil Liability Act. In December 2014 GIC Re was working with the AEC to prepare a proposal for a nuclear insurance pool, with either the building contractor or the operator taking out insurance to cover the suppliers. In January 2015 following President Obama's visit, a Rs 7.5 billion crore ($122 million) nuclear insurance pool was announced by the foreign ministry, with the government to provide more cover for operators "on a tapering basis". The pool would be set up by GIC Re and four other general insurance providers in the public sector (Oriental, New India Assurance, United India and National Insurance). GIC and the government would each contribute Rs 750 crore initially.

In June 2015 the Rs 1,500 crore ($234 million) India Nuclear Insurance Pool (INIP) was announced by GIC Re, which will manage it. The UK pool, Nuclear Risk Insurers, helped to set up INIP and was to be part of the consortium, with 11 domestic insurers, and to provide Rs 500 crore reinsurance, but this NRI involvement did not proceed and the risk is not able to be reinsured under the 2010 Act. One of the consortium members, state-controlled New India Assurance will issue policies and manage coverage for operators and suppliers, initially for third-party liability. GIC Re as a pool manager aims to develop INIP into a one-stop facility for covering all nuclear risks. DAE said the pool should address concerns of suppliers, and GIC Re is planning to launch a complementary product specifically to do that. In January 2016 the cabinet asserted that "international and domestic concerns" over India's liability laws had been resolved with the 2015 establishment of the India Nuclear Insurance Pool, but this is not the perception of suppliers.

Early in 2016 NPCIL faced an equipment sourcing problem for two of its indigenous reactor projects under construction, Kakrapar in Gujarat and Rajasthan in that state, with even domestic supply chain vendors reluctant to provide components, as NPCIL refuses to give them indemnity from the liability provisions of the 2010 Act.

In July 2016 it was reported that the New India Assurance Company had issued a public liability policy to NPCIL covering all its power plants for INR 150 billion ($2.2 billion) for a total premium of about INR 1 billion.

Non-proliferation, US-India agreement and Nuclear Suppliers Group

India's nuclear industry has been largely without IAEA safeguards, though four nuclear power plants (see above) have been under facility-specific arrangements related to India's INFCIRC/66 safeguards agreement with IAEA. However, in October 2009 India's safeguards agreement with the IAEA became operational, with the government confirming that 14 reactors would be put under the India Specific Safeguards Agreement by the end of 2014.

India's situation as a nuclear-armed country excluded it from the Nuclear Non-Proliferation Treaty (NPT)* so this and the related lack of full-scope IAEA safeguards meant that India was isolated from world trade by the Nuclear Suppliers' Group. A clean waiver to the trade embargo was agreed in September 2008 in recognition of the country's impeccable non-proliferation credentials. India has always been scrupulous in ensuring that its weapons material and technology are guarded against commercial or illicit export to other countries.

* India could only join the NPT if it disarmed and joined as a Non Nuclear Weapons State, which is politically impossible. See Appendix.

Following the 2005 agreement between US and Indian heads of state on nuclear energy cooperation, the UK indicated its strong support for greater cooperation and France then Canada then moved in the same direction. The US Department of Commerce, the UK and Canada relaxed controls on export of technology to India, though staying within the Nuclear Suppliers Group guidelines. The French government said it would seek a nuclear cooperation agreement, and Canada agreed to "pursue further opportunities for the development of the peaceful uses of atomic energy" with India.

In December 2006 the US Congress passed legislation to enable nuclear trade with India. Then in July 2007 a nuclear cooperation agreement with India was finalized, opening the way for India's participation in international commerce in nuclear fuel and equipment and requiring India to put most of the country's nuclear power reactors under IAEA safeguards and close down the CIRUS research reactor at the end of 2010. It would allow India to reprocess US-origin and other foreign-sourced nuclear fuel at a new national plant under IAEA safeguards. This would be used for fuel arising from those 14 reactors designated as unambiguously civilian and under full IAEA safeguards.

The IAEA greeted the deal as being "a creative break with the past" – where India was excluded from the NPT. After much delay in India's parliament, it then set up a new and comprehensive safeguards agreement with the IAEA, plus an Additional Protocol. The IAEA board approved this in July 2008, after the agreement had threatened to bring down the Indian government. The agreement is similar to those between IAEA and non nuclear weapons states, notably Infcirc-66, the IAEA's information circular that lays out procedures for applying facility-specific safeguards, hence much more restrictive than many in India's parliament wanted.

The next step in bringing India into the fold was the consensus resolution of the 45-member Nuclear Suppliers Group (NSG) in September 2008 to exempt India from its rule of prohibiting trade with non-members of the NPT. A bilateral trade agreement then went to US Congress for final approval, and was signed into law on 8 October 2008. Similar agreements apply with Russia and France. The ultimate objective is to put India on the same footing as China in respect to responsibilities and trade opportunities, though it has had to accept much tighter international controls than other nuclear-armed countries.

The introduction to India's safeguards agreement says that India's access to assured supplies of fresh fuel is an "essential basis" for New Delhi's acceptance of IAEA safeguards on some of its reactors and that India has a right to take "corrective measures to ensure uninterrupted operation of its civilian nuclear reactors in the event of disruption of foreign fuel supplies." But the introduction also says that India will "provide assurance against withdrawal of safeguarded nuclear material from civilian use at any time." In the course of NSG deliberations India also gave assurances regarding weapons testing.

In October 2008 US Congress passed the bill allowing civil nuclear trade with India, and a nuclear trade agreement was signed with France. The 2008 agreements ended 34 years of trade isolation in relation to nuclear materials and technology. The CIRUS research reactor was shut down on 31 December 2010.

India's safeguards agreement (INFCIRC/754) was signed early in 2009, though the timeframe for bringing the extra reactors (Kakrapar 1&2 and Narora 1&2, beyond Tarapur 1&2, Rawatbhata 1-6 and Kudankulam 1&2) under safeguards still had to be finalized. An Additional Protocol to the safeguards agreement was agreed by the IAEA Board in March and signed in May 2009 by India. The decision to ratify was announced under the new

government in June 2014, with 20 facilities listed, including six at the Nuclear Fuel Complex, Hyderabad and two stores at Tarapur, plus 12 reactors. Narora 1&2 were not listed by then, but were brought under safeguards at the end of 2014, bringing the total to 22 facilities safeguarded. The Additional Protocol came into force on 25 July 2014, giving the IAEA enhanced access to India's civil power facilities, but actually excluding those facilities not listed. It has a long annex covering (non-existent) exports.

Several essentially civil nuclear power reactors, the new 500 MWe fast breeder reactor at Kalpakkam, and the small enrichment plants for naval fuel remain outside IAEA safeguards.

In 2014 a bilateral agreement with Australia was signed, for supply of uranium. After prolonged consideration the Australian parliamentary committee (JSCOT) charged with recommending on this urged caution regarding Australian uranium sales to India. It recommended tightening of concessions granted under the 2007 US nuclear cooperation agreement with India and the 2009 Indian safeguards agreement with the IAEA. In particular it recommended full separation of civil and military facilities (as verified by the IAEA), and setting up an independent nuclear regulator – an initiative which has been stalled since the Indian government announced in 2011 a new independent and autonomous Nuclear Regulatory Authority of India that was to subsume the present regulator (see above). While JSCOT recommended ratifying the bilateral treaty, it said that uranium sales should begin only after these and other conditions concerning routine inspections and reactor decommissioning plans were fulfilled. In addition it recommended publication of legal advice on consent to reprocessing used fuel provisions in the treaty. If the Australian government accepts most of the recommendations and ratifies the treaty, it will put significant pressure on the Indian government to move forward on undertakings given over the last eight years.

In April 2012 India told the UN Security Council that given its ability and willingness to promote global non-proliferation objectives, and that it already adhered to the guidelines of the Nuclear Suppliers Group (NSG) and the Missile Technology Control Regime (MTCR), "as a country with the ability and willingness to promote global non-proliferation objectives, we believe that the next logical step is India's membership of the four export control regimes." The other two 'regimes' are the informal Australia Group (re chemical and biological weapons) and the Wassenaar Arrangement on export control for conventional arms and dual-use goods and technologies. India also supports the early commencement of negotiations in the Conference of Disarmament in Geneva on a

Fissile Material Cut-off Treaty. Following ratification of the Additional Protocol, India will pursue membership of these four export control regimes. In May 2016 India formally applied to join the NSG.

Appendix

BACKGROUND TO NUCLEAR PROLIFERATION ISSUES

India (along with Pakistan and Israel) was originally a 'threshold' country in terms of the international non-proliferation regime, possessing, or quickly capable of assembling one or more nuclear weapons: Their nuclear weapons capability at the technological level was recognized (all have research reactors at least) along with their military ambitions. Then in 1998 India and Pakistan's military capability became more overt. All three remained remained outside the 1970 Nuclear Non-Proliferation Treaty (NPT), which 186 nations have now signed. This led to their being largely excluded from trade in nuclear plant or materials, except for safety-related devices for a few safeguarded facilities.

India is opposed to the NPT as it now stands, since it is excluded as a Nuclear Weapons State, and has consistently criticized this aspect of the Treaty since its inception in 1970.

Regional rivalry

Relations between India and Pakistan are tense and hostile, and the risks of nuclear conflict between them have long been considered quite high.

In 1974 India exploded a "peaceful" nuclear device at Pokhran and then in May 1998 India and Pakistan each exploded several nuclear devices underground. This heightened concerns regarding an arms race between them.

Kashmir is a prime cause of bilateral tension, its sovereignty has been in dispute since 1948. There is persistent low level military conflict due to Pakistan backing a Muslim rebellion there.

Both countries engaged in a conventional arms race in the 1980s, including sophisticated technology and equipment capable of delivering nuclear weapons. In the 1990s the arms race quickened. In 1994 India reversed a four-year trend of reduced allocations for defence, and despite its much smaller economy, Pakistan pushed

its own expenditures yet higher. Both then lost their patrons: India, the former USSR; and Pakistan, the USA.

In 1997 India deployed a medium-range missile and is now developing a long-range missile capable of reaching targets in China's industrial heartland.

In 1995 the USA quietly intervened to head off a proposed nuclear test. The 1998 tests were unambiguously military, including one claimed to be of a sophisticated thermonuclear device. Their declared purpose was "to help in the design of nuclear weapons of different yields and different delivery systems".

It is the growth and modernization of China's nuclear arsenal and its assistance with Pakistan's nuclear power program and, reportedly, with missile technology, which now exacerbates Indian concerns. In particular, China's People's Liberation Army operates somewhat autonomously within Pakistan as an exporter of military material.

Indian security policies are driven by:

- its desire to be recognized as the dominant power in the region;
- its increasing concern with China's expanding nuclear weapons and missile delivery programs; and
- its enduring concern about Pakistan, with its nuclear weapons capability and now the clear capability to deliver such weapons deep into India.

It perceives nuclear weapons as a cost-effective political counter to China's nuclear and conventional weaponry, and the effects of its nuclear weapons policy in provoking Pakistan is, by some accounts, considered incidental.

India has had an unhappy relationship with China. Soundly defeated by China in the 1962 war, relations were frozen until 1998. Since then a degree of high-level contact has been established and a few elementary confidence-building measures put in place. China still occupies some Indian territory. Its nuclear and missile support for Pakistan is however a major bone of contention.

India's weapons material initially came from the Canadian-designed 40 MWt CIRUS "research" reactor which started up in 1960 (well before the NPT), and the 100 MWt Dhruva indigenous unit in operation since 1985, using local uranium. CIRUS was

supplied with heavy water from the USA and it was probably only after the 1962 war that it was employed largely to make weapons-grade plutonium.* Development of nuclear weapons apparently began in earnest in 1967. It is estimated that India may have built up enough weapons-grade plutonium for one hundred nuclear warheads.

* Article III of the 1956 India-Canada Agreement: The Government of India will ensure that the reactor and any products resulting from its use will be employed for peaceful purposes only. Clause 9 of the India-US Heavy Water Agreement: The heavy water sold here under shall be for use only in India by the Government in connection with research into and the use atomic energy for peaceful purposes.

In response to India's 1974 nuclear test explosion using plutonium from CIRUS, demonstrating that nuclear technology transferred to non-nuclear-weapons states for peaceful purposes could be misused, the Nuclear Suppliers Group was formed and began regulating nuclear trade, particularly with India. This is one reason why the closure of CIRUS is a condition of the NSG waiver in 2008.

Nuclear arms control in the region

The public stance of India and Pakistan on non-proliferation differs markedly.

Pakistan has initiated a series of regional security proposals. It has repeatedly proposed a nuclear-free zone in South Asia and has proclaimed its willingness to engage in nuclear disarmament and to sign the NPT if India would do so. This would involve disarming and joining as non-weapon states. It has endorsed a US proposal for a regional five power conference to consider non-proliferation in South Asia.

India has taken the view that solutions to regional security issues should be found at the international rather than the regional level, since its chief concern is with China. It therefore rejects Pakistan's proposals.

Instead, the 'Gandhi Plan', put forward in 1988, proposed the revision of the NPT, which it regards as inherently discriminatory in favour of the Nuclear-Weapons States, and a timetable for complete nuclear weapons disarmament. It endorsed early proposals for a Comprehensive Test Ban Treaty (CTBT) and for an international convention to ban the production of highly enriched

uranium and plutonium for weapons purposes, known as the 'cut-off' convention.

The USA has, for some years pursued a variety of initiatives to persuade India and Pakistan to abandon their nuclear weapons programs and to accept comprehensive international safeguards on all their nuclear activities. To this end the Clinton administration proposed a conference of nine states, comprising the five established nuclear-weapon states, along with Japan, Germany, India and Pakistan.

This and previous similar proposals have been rejected by India, which countered with demands that other potential weapons states, such as Iran and North Korea, should be invited, and that regional limitations would only be acceptable if they were accepted equally by China. The USA would not accept the participation of Iran and North Korea and such initiatives lapsed.

Another, more recent approach, centres on the concept of containment, designed to 'cap' the production of fissile material for weapons purposes, which would hopefully be followed by 'roll back'. To this end India and the USA jointly sponsored a UN General Assembly resolution in 1993 calling for negotiations for a 'cut-off' convention, the Fissile Material Cut-off Treaty (FMCT). Should India and Pakistan join such a convention, they would have to agree to halt the production of fissile materials for weapons and to accept international verification on their relevant nuclear facilities (enrichment and reprocessing). In short, their weapons programs would be thus 'capped'. It appeared that India was prepared to join negotiations regarding such a FMCT under the 1995 UN Conference on Disarmament (UNCD).

However, despite the widespread international support for a FMCT, formal negotiations on cut-off have yet to begin. The UNCD can only approve decisions by consensus and since the summer of 1995, the insistence of a few states to link FMCT negotiations to other nuclear disarmament issues has brought progress on the cut-off treaty there to a standstill. In connection with its 2006 agreement with the USA, India has reiterated its support for a FMCT.

Bilateral confidence-building measures between India and Pakistan to reduce the prospects of confrontation have been limited. In 1990 each side ratified a treaty not to attack the other's nuclear installations, and at the end of 1991 they provided one another with a list showing the location of all their nuclear plants, even though the respective lists were regarded as not being wholly

accurate. Early in 1994 India proposed a bilateral agreement for a 'no first use' of nuclear weapons and an extension of the 'no attack' treaty to cover civilian and industrial targets as well as nuclear installations.

Having promoted the CTBT since 1954, India dropped its support in 1995 and in 1996 attempted to block the Treaty. Following the 1998 tests the question has been reopened and both Pakistan and India have indicated their intention to sign the CTBT. Indian ratification may be conditional upon the five weapons states agreeing to specific reductions in nuclear arsenals.

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